1 HI-Jass User Manual

1.1 Running the application

1.1.1 Linux

```
./run_hi_jass.sh
```

1.1.2 Windows

Double-click `run_hi_jass.bat`, or run:

```
.\run_hi_jass.bat
```

The launcher creates the virtual environment, installs the dependencies, and starts the GUI.

1.2 Default NBI configuration

The default configuration contains two neutral beams:

Beam	Species	Injection energy	Injected power	Tangent R_t	Direction
NBI-1	D	120 keV	10 MW	0.5 m	co
NBI-2	T	180 keV	0.1 MW	0.6 m	counter

1.3 Beam vertical angle

`BeamParams.vertical_angle_deg` (α, HOT-Jass: NBI "vertical angle", default 0) inclines the beam axis to its horizontal projection about the fixed tangency point (R_t, Z_t); $\alpha > 0$ rises along the travel. With y the horizontal distance from the tangency point (negative before it):

$$R(y) = \sqrt{R_t^2 + y^2}, \qquad Z(y) = Z_t + y \tan\alpha, \qquad ds = dy/\cos\alpha.$$

`physics.tangential_chords` finds the entry and exit through the LCFS by bisection on the outermost crossings (samples in between outside the plasma, e.g. in the hole, carry $n_e = 0$); the centre-post blocks at the same y as for $\alpha = 0$. Shine-through, the birth profile and the first-orbit-loss chord all use this path, and the vertical ray offsets of a finite beam become $v/\cos\alpha$ in Z. The birth pitch (below) includes the inclination, and the NBI torque is multiplied by $\cos\alpha$, since only the horizontal velocity carries toroidal momentum. $\alpha = 0$ keeps the horizontal-chord geometry exactly.

1.4 Field directions and the beam birth pitch

Convention: $I_p > 0$ and $B_t > 0$ both counter-clockwise seen from above ($+\phi$ in right-handed (R, ϕ, Z)), as the HOT-Jass Geometry arrows show; B is along I_p. In the poloidal (R, Z) view $+\phi$ points into the page and $\mathbf{B}_p = B_p\, \hat{e}_\phi \times \hat{n}$ ($\hat{n}$ the outward normal of the elliptical flux surface) circulates clockwise: outward at the top, down on the outboard side.

Reversed fields: `PlasmaParams.ip_sign` / `bt_sign = -1` (HOT-Jass: negative Ip / B0 inputs) reverse I_p (and $\mathbf{B}_p$ with it) or B_t. The plasma current and field stay magnitudes in every scaling, q, orbit-width and Greenwald formula; the signs enter only the birth pitch, and co-current beams

always run along I_p. The pitch is returned relative to I_p ($\xi_0 = \sigma_I \sigma_B \xi_{lab}$), so only the relative direction of B_t and I_p matters: reversing I_p alone is the mirror image of the machine in a vertical plane (identical results), reversing both is the machine turned upside down (up- and down-inclined beams, and $\pm Z_t$, swap), and reversing B_t alone is therefore equivalent to swapping the up and down inclinations.

The birth pitch is taken against the full field (`physics.beam_birth_pitch`),

$$\xi_0 = \frac{\mathbf{v}\cdot\mathbf{B}}{vB} = \frac{\cos\alpha\,(y/R)\,B_R \pm \cos\alpha\,(R_t/R)\,B_\phi + \sin\alpha\,B_Z}{\sqrt{B_\phi^2 + B_p^2}}, \qquad B_\phi = B_t R_0/R,$$

$\pm$ for co / counter beams, with $B_p(\rho) = \mu_0 I(\rho)/L_p(\rho)$ from the assumed current profile $j \propto (1-\rho^2)^\nu$, $\nu = q_{cyl}/q_0 - 1$ (`physics.poloidal_field_T`, the same profile as the Current tab; `TokamakConfig.q0`). The B_R and B_Z terms make an up- and a down-inclined beam differ, and the normalisation by $|B|$ lowers $|\xi_0|$ below the toroidal direction cosine R_t/R by B_ϕ/B. ξ_0 is used by the pitch-resolved ST orbit loss (`first_orbit_loss_fraction_st(..., pitch=)`), the consistent birth weighting and NBCD; an ion born with $\xi_0 < 0$ moves against I_p and is treated as counter-current.

HOT-Jass web, Beam tab: the *Fast-ion birth vs normalised radius* panel also plots each NBI's birth-weighted mean pitch $\xi_0(\rho)$ for its full-energy component (dash-dot, right axis), from the same deposition model (`beam_birth_density_profile(..., return_pitch=True)`, with q_0 and the I_p/B_t signs the solver uses); it is blanked where the beam deposits less than 2 % of its peak.

1.5 Beam energy components (full, half, third)

A positive-ion source also extracts D2+ and D3+ (H2+/H3+, T2+/T3+), which dissociate in the neutraliser into atoms at $E_b/2$ and $E_b/3$. Each beam therefore carries **power fractions** f_1, f_2, f_3 of its full-, half- and third-energy components (`BeamParams.energy_fractions`, default $(1, 0, 0)$ – a mono-energetic beam):

$$E_k = E_b/k, \qquad P_k = f_k\,P_{NB}, \qquad k = 1, 2, 3.$$

`HotJassModel` splits every beam into its non-zero components before the solve (`beam_components()`), so shine-through, orbit and CX losses, slowing-down, beam-target fusion and current drive all run per component, and the per-beam result lists (`f_capture`, `I_nb_per_beam_A`, ...) hold one entry per component. The particle fractions follow from $N_k \propto P_k/E_k$, i.e. $\propto k f_k$. Beam-beam fusion sums over every component pair of the two beamlines.

1.6 Two-beam slowing-down model

The slowing-down quantities are calculated independently for each beam. For beam j:

$$E_{c,j} = \left(\frac{m_j}{m_e}\right)^{1/3} T_e$$

$$\tau_{se,j} = 6.27 \times 10^8\,\frac{A_j T_e[\mathrm{eV}]^{3/2}}{n_e[\mathrm{cm}^{-3}]\ln\Lambda}$$

$$\tau_{S,j} = \frac{\tau_{se,j}}{3}\ln\left[1 + \left(\frac{E_{b,j}}{E_{c,j}}\right)^{3/2}\right]$$

The useful beam power includes the HotJass capture and loss cascade. The fast-ion density from each beam is:

$$n_{b0,j} = \frac{P_{\text{useful},j}\tau_{S,j}}{E_{b,j}(10^3 e)V}$$

The total fast-ion density is:

$$n_{b0} = \sum_j n_{b0,j}$$

1.7 Density peaking and shine-through

The density peaking input defines the analytic electron-density profile:

$$n_e(\rho) = n_{e0}(1-\rho^2)^{2p_n}, \qquad 0 \le \rho \le 1$$

Shine-through uses the density along the beam chord rather than assuming that the central density applies everywhere. Each NBI supplies a tangent point (R_t, Z_t) in the NBI tab. The straight tangential chord at that point is intersected with the shaped plasma cross-section, and the optical depth is evaluated as:

$$\tau_b = \sigma_{\text{stop}}(E_b/A_b)\int_{\text{chord}} n_e(\rho(s))\,ds$$

and the captured and shine-through fractions are:

$$f_{\text{capt},b} = 1 - e^{-\tau_b}, \qquad f_{\text{shine},b} = e^{-\tau_b}.$$

The chord is evaluated numerically using the local cylindrical radius along the tangent path and the plasma elongation. With $p_n = 0$, the density is uniform along the computed chord; with $p_n > 0$, the profile is integrated point by point. A tangent point outside the plasma produces zero captured power.

1.7.1 Shine-through model selection

Each NBI has a three-way shine-through selector:

- **Riviere**: the existing Riviere-type analytic stopping cross-section fit.
- **Janev/Suzuki**: the Janev-Boley-Post analytic fit from the supplied paper, Nuclear Fusion 29 (1989) 2125, Eqs. (23)-(26), using the published Table III coefficients and the carbon-impurity coefficients from Table IV. It models total effective stopping, including the paper's density, electron-temperature and Z_{eff} dependence. The fit is published for $100 \le E \le 10^4$ keV/u, $10^{12} \le n_e \le 10^{15}$ cm^{-3} and $1 \le T_e \le 50$ keV; inputs outside these ranges are clipped to the nearest validity boundary.
- **Manual**: the entered manual shine-through fraction is used directly; captured power is $(1 - f_{\text{shine}})P_{NB}$.

The selected calculated stopping model is also used by the optional first-orbit-loss attenuation weighting. The Manual mode bypasses the cross-section calculation for shine-through.

1.8 First-orbit loss (reduced model)

First-orbit loss is an **opt-in** channel (the *First-orbit loss* checkbox in the Losses section). When enabled, it estimates the fraction of *captured* fast ions that leave the plasma promptly on their first drift orbit, before any slowing-down or charge-exchange because they are born close enough to the last closed flux surface (LCFS) that their guiding-centre orbit, or their gyro-orbit, crosses it.

The **Orbit model** selector (Losses section) chooses how the orbit widths and the loss criterion are built:

- **Large-aspect ($q_*\rho_{Li}$)** the original reduced model described in *Passing-orbit width* and *Loss criterion* below. Direction-only mean shift; exactly zero co-current loss.
- **ST orbits mean-shift (arbitrary A)** and **ST orbits pitch-resolved** two spherical-tokamak variants described in *ST orbit models* at the end of this section. Both give non-zero co-current loss.
- **ST orbits $\mathbf{P}_\phi$ orbit boundary** births followed along their guiding-centre orbits from the constants of motion (*Orbit-boundary model* at the end of this section). In HOT-Jass web it is the default of the spherical presets (PONTE, ST40, MAST-U, NSTX-U); JET and TCV default to the pitch-resolved model, ITER and T-15MD to the large-aspect one.

1.8.1 Birth profile along the chord

The calculation reuses the shine-through deposition. Along the straight tangential chord (assumed $R_t = R_0$, length $2a$) the local ionisation rate density is

$$\dot{n}(x) \propto n_{e0}\,\sigma_{\mathrm{stop}}(E_{b,j}/A_j)\,e^{-n_{e0}\sigma_{\mathrm{stop}}x}, \qquad x \in [0, 2a],$$

and the path coordinate is mapped to a normalised flux label by

$$\rho(x) = \frac{|x - a|}{a},$$

so $\rho = 0$ at the chord mid-point and $\rho = 1$ at both ends. The stopping cross-section is the beam's selected shine-through model; **Manual** mode has no deposition shape, so the Riviere fit is substituted for this weighting only.

Off-axis / vertically-shifted injection. When a beam's tangency radius R_t differs from R_0, or its aim height $Z_t \neq 0$, the birth profile is built instead from the *real* chord: a horizontal line tangent to the cylinder $R = R_t$ at height Z_t, sampled through elliptical flux surfaces

$$\rho(s)^2 = \left(\frac{R(s) - R_0}{a}\right)^2 + \left(\frac{Z_t}{\kappa a}\right)^2, \qquad R(s) = \sqrt{R_t^2 + (s - s_0)^2},$$

with the chord length running edge-to-edge and ρ no longer reaching 0 (the beam no longer passes through the magnetic axis). This is the same geometry the shine-through / capture calculation already uses. For the standard on-axis tangential case ($Z_t = 0$, $R_t = R_0$ or unset) the crude $\rho(x) = |x - a|/a$ mapping above is kept, so existing results are unchanged.

Central column. The beam enters from the outboard side. If its tangency radius is inside the central column ($R_t < R_{\mathrm{post}}$, set by `Centre-post R`; default $R_0 - a$), the straight chord would strike the column, so only the **outboard leg** from the outboard-edge entry to the point where $R(s) = R_{\mathrm{post}}$ is kept; the (unreachable) far leg is discarded. A beam with R_t at or outboard of

R_{post} threads the gap between the column and the inboard plasma edge and keeps its full chord. A beam whose aim point lies outside the plasma altogether contributes no first-orbit loss. The Deposition tab's geometry panel draws the column and marks a blocked beam.

1.8.2 Passing-orbit width

The radial width of a strongly co-passing drift orbit at the full injection energy is the textbook large-aspect-ratio estimate

$$\Delta r_j = q_*\,\rho_{Li,j}, \qquad q_* = \frac{2\pi a^2 B_t}{\mu_0 R_0 I_p}\,\frac{1+\kappa^2}{2}, \qquad \rho_{Li,j} = \frac{m_j v_{b,j}}{eB_t}.$$

q_* is the cylindrical, elongation-corrected safety factor and $\rho_{Li,j}$ the toroidal-field gyroradius at $E_{b,j}$.

1.8.3 Loss criterion

The orbit is modelled as a **rigid mean shift** of the birth radius: inward for co-current injection (favourable), outward for counter-current:

$$\rho_{\mathrm{eff}}(x) = \rho(x) \mp \frac{\Delta r_j}{a} \qquad (\mathrm{co}\,/\,\mathrm{counter}),$$

and an ion born at x is counted as lost if $\rho_{\mathrm{eff}}(x) > 1$. Because $\rho(x) \le 1$, the co-current shift can never trigger this test, so the mean-shift model gives **exactly zero first-orbit loss for co-current injection**; for counter-current it loses every ion born with $\rho(x) > 1 - \Delta r_j/a$.

An optional, direction-independent **Larmor (gyro-orbit) prompt-loss** channel can be OR-ed in: an ion whose guiding centre is born within one gyro-diameter of the LCFS,

$$\rho(x) > 1 - \frac{2\rho_{Li,j}}{a},$$

is lost regardless of the drift direction (gyration is much faster than the guiding-centre drift). This channel is disabled by default.

1.8.4 Loss fraction and power

$$f_{\mathrm{orbit},j} = \frac{\int_{\mathrm{lost}\ x} \dot{n}(x)\,dx}{\int_0^{2a} \dot{n}(x)\,dx}, \qquad P_{\mathrm{orbit},j} = f_{\mathrm{orbit},j}\,P_{\mathrm{capt},j}.$$

First-orbit loss is applied to the captured power *before* the charge-exchange loss and before the useful power that drives T_e, T_i and $n_{b0,j}$. Injection direction (co-/counter-current) is set per NBI in its input section.

1.8.5 Reported quantities

The Dashboard and the Assumptions tab report, per beam, $\rho_{Li,j}$ and Δr_j (in cm and as fractions of a) and $f_{\mathrm{orbit},j}$.

1.8.6 Limitations

- **The co-current zero is a model artefact, not physics.** The rigid mean-shift ignores the finite radial width of the orbit and the outboard excursion that a favourable inward mean shift does not cancel; published orbit-following comparisons give of order a few percent co-current first-orbit loss, versus tens of percent counter-current.
- The chord label $\rho(x) = |x - a|/a$ folds the inboard and outboard halves together, so the model cannot distinguish a high-field-side birth from a low-field-side birth. It therefore cannot represent the correct structure co-current losing high-field-side edge births, counter-current losing low-field-side edge births, and trapped-ion banana-tip losses on the low-field side in either direction.
- Circular / elliptical cross-section, large-aspect-ratio orbit widths (the birth chord itself now honours R_t and Z_t, but the orbit-width and pitch estimates are still evaluated at R_0); Δr_j and $\rho_{Li,j}$ are evaluated at the full injection energy (an upper bound real ions slow down).
- The birth-profile integral in this channel uses the flat central density n_{e0}, not the peaked profile used elsewhere.
- When ρ_{Li}/a or $\Delta r/a \gtrsim 0.3$ orbit width comparable to the machine size, typical of compact low-aspect-ratio devices the whole estimate is order-unity uncertain and really needs a guiding-centre orbit follower. The Assumptions tab flags this case. The ST orbit models below address the first three of these limitations directly.

1.8.7 ST orbit models (arbitrary aspect ratio)

On a spherical tokamak the drift and banana orbit widths are an $O(a)$ fraction of the minor radius, the trapped fraction is large, and outboard-born trapped ions are lost **regardless of injection direction** so the large-aspect model's exact co-current zero is qualitatively wrong. The two ST variants build their widths from the **poloidal gyroradius**

$$\rho_{\theta,j} = \frac{B_t}{\bar{B}_p}\,\rho_{Li,j}, \qquad \bar{B}_p = \frac{\mu_0 I_p}{2\pi a\sqrt{(1+\kappa^2)/2}},$$

(not $q_*\rho_{Li}$, whose q_* runs away as $\varepsilon \to 1$), together with the circulating and banana widths and the arbitrary-A trapped fraction

$$w_{\mathrm{pass},j} = \varepsilon\,\rho_{\theta,j}, \qquad w_{\mathrm{ban},j} = 2\sqrt{\varepsilon}\,\rho_{\theta,j}, \qquad f_t = 1 - \frac{(1-\varepsilon)^2}{(1+1.46\sqrt{\varepsilon})\sqrt{1-\varepsilon^2}},$$

with $\varepsilon = a/R_0$ (clamped to $[10^{-3}, 0.95]$) and $f_c = 1 - f_t$ (Lin-Liu & Miller, *Phys. Plasmas* **2** (1995) 1666). An arbitrary-A edge safety factor q_a (Uckan / ITER Physics Basis form, with the $(1.17 - 0.65\varepsilon)/(1-\varepsilon^2)^2$ shaping factor) is reported for reference but is **not** used in the widths.

The birth profile in this channel uses the same Beer-Lambert deposition density as the large-aspect model and the Deposition tab's chord plot, $n_e\sigma\, e^{-n_e\sigma x}$ via the selected stopping model so it responds to n_{e0} the same way everywhere in the code (an earlier version used a fixed "penetration rule of thumb" mean free path that saturated at its clamp for realistic densities, decoupling the ST models' loss fraction from n_{e0}; this has been fixed).

The **drift-orbit shift is direction-dependent** for both populations inward for co-current ($v_\parallel$ along I_p), outward for counter with a weaker shift $c_t\, w_{\mathrm{pass}}$ ($c_t = 0.25$) for the trapped-ion banana centre than for a passing ion. The **gyro channel** ($\rho > 1 - \rho_{Li}/a$, born within one gyroradius of the LCFS) is direction-independent. Co-current loss is therefore non-zero but smaller than counter-current the co/counter asymmetry survives, unlike the large-aspect model's all-or-nothing 0 vs 1.

Mean-shift variant. Deposition-weighted loss integrals over $\rho(x)$, blended by f_c, f_t (sign convention: − co, + counter):

$$f_{\mathrm{orbit},j} = f_c\, L_{\mathrm{pass}} + f_t\, L_{\mathrm{trap}},$$

$$L_{\mathrm{pass}}:\ \rho \mp \frac{w_{\mathrm{pass},j}}{a} > 1 \ \text{ or } \ \rho > 1 - \frac{\rho_{Li,j}}{a},$$

$$L_{\mathrm{trap}}:\ \rho + \frac{w_{\mathrm{ban},j}}{2a} \mp \frac{0.25\, w_{\mathrm{pass},j}}{a} > 1 \ \text{ or } \ \rho > 1 - \frac{\rho_{Li,j}}{a}.$$

Pitch-resolved variant. A 2-D integral over birth radius x and the pitch **magnitude** $|\lambda| = |v_\parallel/v|$. The birth value $|\lambda_0(x)| = R_t/\sqrt{R_t^2 + (x-a)^2}$ is fixed by the injection *geometry* and does **not** depend on the poloidal-field / current direction only the sign of $v_\parallel$ relative to I_p flips (co ↔ counter), carried by $s_{\mathrm{dir}} = \mp 1$. $|\lambda_0|$ is smeared by a $\sigma = 0.10$ Gaussian. An ion is trapped if $|\lambda| < \sqrt{2\varepsilon_{\mathrm{loc}}/(1+\varepsilon_{\mathrm{loc}})}$ ($\varepsilon_{\mathrm{loc}} = \rho\, a/R_0$); its outboard radial excursion is

$$\text{passing}:\ s_{\mathrm{dir}}\, \frac{w_{\mathrm{pass},j}}{a}\, |\lambda|, \qquad \text{trapped}:\ \frac{w_{\mathrm{ban},j}}{2a}\, \frac{|\lambda|}{\lambda_{\mathrm{tr}}} + s_{\mathrm{dir}}\, \frac{0.25\, w_{\mathrm{pass},j}}{a},$$

(the banana half-width → 0 for deeply trapped ions, largest near the trapped/passing boundary), and it is lost if ρ + (excursion) > 1 or $\rho > 1 - \rho_{Li,j}/a$.

Both variants collapse toward the large-aspect co-current zero as $\varepsilon \to 0$. They remain 0-D order-of-magnitude estimates absolute magnitudes are upper-bound-like and, because the direction-independent trapped banana-tip and gyro channels stay significant, the co/counter split is narrower than the large-aspect model's (typically a factor ~ 1.5, not 0 vs 1). References: Akers *et al.*, *Nucl. Fusion* (START NBI, $A \sim 1.4$); Goldston, White & Boozer, *Phys. Rev. Lett.* **47** (1981) 1004; Goldston & Rutherford, *Introduction to Plasma Physics* (1995), Ch. 12.

1.8.8 Orbit-boundary model from the constants of motion (`orbit_model = "st_pphi"`)

HOT-Jass: MODELS, "ST orbits - P_ϕ orbit boundary". Instead of estimating drift and banana widths, each birth is followed along its guiding-centre orbit (`physics.pphi_orbit_loss_probability`). Energy, magnetic moment $\mu = mv_\perp^2/2B$ and canonical toroidal angular momentum

$$P_\phi = mRv_\phi - e\Psi$$

are conserved, with $\Psi(\rho)$ the poloidal flux per radian, increasing outward relative to I_p, $d\Psi/d\rho = R_0 B_p(\rho)\, a\sqrt{(1+\kappa^2)/2}$ (`physics.poloidal_flux_profile`, B_p from the assumed current profile, see "Field directions and the beam birth pitch"). With $B \approx B_0 R_0/R$ and $v_\phi \approx v_\parallel$, a birth at (R_b, Ψ_b) with signed pitch ξ_0 (positive along I_p) sits at major radius R of an orbit leg on

$$\Psi(R) = \Psi_b + \frac{m}{e}\left[R\, v_\parallel(R) - R_b v\, \xi_0\right], \qquad v_\parallel(R) = \pm v\sqrt{1 - (1-\xi_0^2)\, R_b/R},$$

with the bounce point at $R = R_b(1-\xi_0^2)$. The orbit exists at R where $\Psi(R) \geq \Psi(\rho = |R - R_0|/a)$ (the flux surface reaches that R). The birth leg is followed over the connected range of R around R_b where it exists; if that range reaches the bounce point (a trapped ion), the return leg (opposite sign of $v_\parallel$) is added. The ion is lost if the outermost flux reached corresponds to $\rho \geq 1 - \rho_{Li}/a$ (the gyro criterion of the other ST models). There are no fitted constants: a co-current ion born on the outboard side is at the outermost point of its orbit (its banana or drift orbit lies inward)

and is lost only within a Larmor radius of the edge; a counter-current ion is at the innermost point, its orbit reaching a full banana or drift width outward. The birth pitch carries the same 0.1 Gaussian spread (beam divergence, finite chord) as the pitch-resolved model, as a 3-point Gauss-Hermite average.

Implementation: in this up-down symmetric approximation the outward excursion $\rho_{out} - \rho_b$ depends on (R_b, ρ_b, ξ_0) only; it is tabulated once per beam energy and machine on a $33 \times 33 \times 65$ grid (`_pphi_excursion_table`, cached, ˜0.2 s) and interpolated, and the loss criterion is applied per birth, which keeps the loss boundary sharp (table against direct orbits: mean loss within ˜2 %). The lost power fraction is the loss probability averaged over the births along the beam-axis chord, weighted by the birth rate $n_e(\rho)\sigma_s e^{-\tau}$ with the real density profile; the birth profile is always weighted by the same probability (no cutoff radius, the "consistent" option is implied).

Approximations: up-down symmetric elliptical flux surfaces; $|B| \approx B_\phi$ in the bounce condition and $v_\phi \approx v_\parallel$; flux-surface averaged B_p ($|\nabla\psi| \approx R_0 B_p$) from the assumed current profile; no wall or centre-post shape beyond the LCFS; collisions during the first orbit neglected.

Comparison with the pitch-resolved model (presets, beam-axis NBI-1 / NBI-2): co-current losses fall strongly (PONTE 23 % → 9 %, MAST-U 30 % → 15 %, NSTX-U 29 % → 19 %, TCV 24 % → 5 %, JET 4.7 % → 2.4 %, T-15MD 5.4 % → 1.4 %), counter-current ones stay large (PONTE NBI-2 55 %, TCV NBI-2 28 %).

References: Rome & Peng, *Nucl. Fusion* **19** (1979) 1193 (orbit topology and loss boundaries from the constants of motion); Eriksson & Porcelli, *Plasma Phys. Control. Fusion* **43** (2001) R145 (review of energetic-ion orbits); Heidbrink & Sadler, *Nucl. Fusion* **34** (1994) 535 (fast ions in tokamak experiments, prompt losses); White, *The Theory of Toroidally Confined Plasmas* (guiding-centre motion, canonical momentum).

1.9 Charge-exchange loss

A captured fast ion can charge-exchange with a background cold neutral during slowing-down, becoming a fast **neutral** that escapes the plasma before fully thermalizing a second loss channel, applied to the post-orbit-loss captured power (Step 0, same ordering as first-orbit loss above). Design note: `docs/CX model.pdf`. A **CX-loss model** selector (rail, "Losses" section) picks between three modes; only the active mode's inputs matter.

1.9.1 Manual fraction (default)

The original flat efficiency knob `cx_loss_fraction` $\in [0, 1]$: a fixed fraction of captured power is removed, independent of E_b, n_e, or any background-neutral density. PPPL-1280 Sect. 6.2.2/6.2.3 quotes "up to 10%" of injected power for $Z_{\text{eff}} = 1$ as a rough qualitative figure, not a formula. Default 0 =off; leaves every other result byte-identical.

1.9.2 Manual n0/ne the physics-based survival integral

The user supplies n_0/n_e directly (a spatially uniform, 0-D background neutral density $n_0 = (n_0/n_e)\, n_{e0}$), and the model evaluates the actual CX survival integral along the slowing-down path. For a fast ion at energy E:

$$\nu_{\text{cx}}(E) = n_0\, \sigma_{\text{cx}}(E)\, v(E), \qquad S(E) = \exp\left[- \int_E^{E_b} \frac{\nu_{\text{cx}}(E')}{|dE'/dt|}\, dE' \right],$$

with particle- and power-loss fractions

$$f_{\mathrm{cx},N} = 1 - S(0^+), \qquad f_{\mathrm{cx},P} = \int_0^{E_b} \frac{E}{E_b}\,\frac{\nu_{\mathrm{cx}}(E)}{|dE/dt|}\,S(E)\,dE.$$

$f_{\mathrm{cx},P}$ (the quantity actually applied to P_{useful}) is smaller than $f_{\mathrm{cx},N}$ because σ_{cx} falls steeply above $\sim$30 to 50 keV/amu, so most CX events happen late in the cascade, when little energy is left to remove. $1/|dE/dt| = (\tau_{se}/2)\sqrt{E}/(E^{3/2} + E_c^{3/2})$ is exactly the same slowing-down kernel already used by `thermalization_time/slowing_down_distribution` (differentiating $\tau_s(E) = (\tau_{se}/3)\ln[1 + (E/E_c)^{3/2}]$ reproduces it), so no new slowing-down physics is introduced only the extra $\nu_{\mathrm{cx}}(E)$ factor under the same integral. $\sigma_{\mathrm{cx}}(E)$ is the Janev & Smith (1993, Sect. 2.3.1) analytic form, refit by Swaczyna, Bzowski & Kubiak (arXiv:2411.13174, 2024, their Eq. A1) against Schultz *et al.* (2023) theory and the Barnett (1990) tabulation quoted accuracy 5% from 1100 keV/amu, the range this project's beams live in.

Also reported, per beam, on the Dashboard: the dimensionless figure of merit $\gamma_{\mathrm{cx}} \equiv \nu_{\mathrm{cx}}(E_b)\,\tau_s \approx n_0\sigma_{\mathrm{cx}}(E_b)v_b\tau_s$ negligible for $\gamma_{\mathrm{cx}} \ll 1$, worth taking seriously for $\gamma_{\mathrm{cx}} \sim 0.1$ to 1, and a sign the single-pass loss treatment is breaking down for $\gamma_{\mathrm{cx}} \gtrsim 1$ (flagged with a warning, as is $f_{\mathrm{cx},P} > 10\%$, PPPL-1280's rough $Z_{\mathrm{eff}} = 1$ ceiling). Because CX loss depends on T_e (via $E_c(T_e)$) while T_e itself depends on the CX-reduced P_{useful}, this is a genuine fixed point; an outer under-relaxed Picard loop (in `hotjass_core`, mirroring the existing alpha-heating fixed point) closes it, keeping the core solver a single non-iterative solve.

1.9.3 Penetration (n0_LCFS/ne) edge-penetration estimate

Instead of a single uniform n_0, the boundary value $n_{0,\mathrm{LCFS}} = (n_{0,\mathrm{LCFS}}/n_e)\,n_{e,\mathrm{LCFS}}$ is set from the user ratio, and the core profile follows the standard penetration/recycling picture. $n_{e,\mathrm{LCFS}}$ is evaluated at $\rho = 0.95$, not literally at $\rho = 1$: the core density shape $n_e(\rho) = n_{e0}(1 - \rho^2)^{2p_n}$ is an exact zero at $\rho = 1$ whenever $p_n > 0$, which would make $n_{0,\mathrm{LCFS}} = 0$ regardless of the input ratio – a knob that does nothing. $\rho = 0.95$ is a compromise: with a strongly peaked profile ($p_n \gtrsim 1$ to 2) it can still sit orders of magnitude below n_{e0}, so treat this mode as unreliable for very peaked density profiles.

$$\lambda_0(\rho) = \frac{v_0}{n_e(\rho)\,[\langle\sigma v\rangle_{\mathrm{ion}}(T_e) + \langle\sigma v\rangle_{\mathrm{cx}}(T_i)]}, \qquad n_0(\rho) \approx n_{0,\mathrm{LCFS}}\exp\left[-\int_\rho^1 \frac{a}{\lambda_0(\rho')}\,d\rho'\right],$$

volume-averaged to a single $\langle n_0\rangle$ and fed through the same survival integral as "Manual n0/ne" above. v_0 is a round ~ 3 eV Franck-Condon-like dissociation speed. **Caveat:** the ionization ($\langle\sigma v\rangle_{\mathrm{ion}}(T_e)$) and thermal-CX ($\langle\sigma v\rangle_{\mathrm{cx}}(T_i)$) rate coefficients used for λ_0 are an interim hand-built approximation (a threshold-shaped form for ionization, a threshold-free $\sqrt{T}$ form for CX, each capped to saturate at a few $\times 10^{-14}\,\mathrm{m}^3/\mathrm{s}$) a calibrated Voronov (1997) / ADAS `adf11` fit was not sourced/verified in this session, so treat $n_0(\rho)$'s absolute scale as order-of-magnitude only; the survival-integral physics downstream of $\langle n_0\rangle$ (same $\sigma_{\mathrm{cx}}(E)$ fit as "Manual n0/ne") is not affected by this caveat. $n_0(\rho)$ itself (per the active mode – flat for "manual n0/ne", the profile above for "penetration", nothing for "manual fraction") is plotted on the Profiles tab; because λ_0 is typically much smaller than a, the profile can span many orders of magnitude edge-to-core (shown on a log axis) – a genuine, if extreme, feature of this reduced model, not a numerical artifact.

1.9.4 Escape probability (informational only)

A fast neutral born from a CX event can re-ionize before reaching the wall, so treating every CX event as a full loss is a conservative upper bound, not exact. An escape-probability factor

$\exp(-x_{\rm wall}/\lambda_{\rm reion})$ is shown per beam (Dashboard) $x_{\rm wall} \approx a$, $\lambda_{\rm reion}$ from the same combined ionization+CX stopping cross section a beam neutral itself uses but is **not** applied to $f_{{\rm cx},P}$; it is informational, per the design note's explicit "acceptable as the conservative default, but shown as such."

1.10 Bulk toroidal rotation

A **Rotation model** selector (Losses section, below the CX-loss fields) picks between three modes, off by default – like every other loss/model knob, this leaves every existing result byte-identical unless explicitly enabled.

Where rotation matters and where it doesn't. v_ϕ (typically $\sim 10^2$ to 10^3 km/s) is small next to the beam's own velocity (~ 3 to 5×10^3 km/s for a 100180 keV D/T beam), so shine-through, first-orbit loss and CX loss all governed by the beam's own velocity are essentially insensitive to it and are **not** touched. The one channel where it matters cleanly is **beam-target fusion**: that reaction's cross-section and flux factor depend on the beam-target *relative* velocity, which the un-rotated model implicitly assumes equals the beam's lab velocity (a stationary target). With rotation:

$$v_{\rm rel}(E) = |\, v_{\rm beam}(E) - s\, v_\phi \,|\,, \qquad s = +1 \text{ (co-current beam)},\ -1 \text{ (counter)},$$

used for *both* the flux factor and the cross-section's energy argument (re-expressed as the equivalent deuteron energy $E_D = \frac{1}{2} M_D v_{\rm rel}^2$, the same conversion `beam_target_reactivity_spectrum()` already used for a T beam). $v_\phi = 0$ reproduces the original stationary-target result exactly.

Non-monotonic, not just a small correction in one direction. The D-T cross-section peaks around ~ 64 keV (deuteron-equivalent energy) and falls off on both sides. For a beam energy *above* that peak (this project's 120180 keV presets all are), pulling $v_{\rm rel}$ down (co-current beam, co-rotating plasma) moves the *high-energy tail* of the slowing-down spectrum back toward the peak and can *increase* beam-target yield, while a counter-current beam sees the opposite – the sign is not "co always suppresses fusion", it depends on where E_b sits relative to the cross-section peak, and the net effect is an integral over the whole slowing-down spectrum (not just E_b itself), evaluated numerically.

1.10.1 Manual v_ϕ

Direct input of a single, spatially-uniform v_ϕ (m/s, positive = co-current) – the honest option when the collisionality of the source is unknown but a plausible rotation speed is.

1.10.2 Momentum balance

v_ϕ instead comes from a 0-D toroidal angular-momentum balance, the rotation analogue of this project's energy balance:

$$T_{\rm NBI} = \sum_j s_j \, \frac{2 P_{{\rm useful},j}\, R_{t,j}}{v_{b,j}}, \qquad v_\phi = \frac{T_{\rm NBI}\, \tau_\phi}{\rho_i\, R_0\, V},$$

summing each beam's injected canonical angular momentum ($s_j = +1$ co, -1 counter; $R_{t,j}$ its own tangency radius), balanced against a momentum-confinement loss L/τ_ϕ with $L \sim \rho_i v_\phi R_0 V$ (ρ_i = thermal-ion mass density, $n_{D0} M_D + n_{T0} M_T$) the bulk plasma treated as a single representative rotating mass at the magnetic axis, the same level of 0-D reduction as everywhere else

in this model. **Caveat:** unlike τ_E, there is no widely validated scaling for the momentum confinement time τ_ϕ; the user-set ratio `tau_phi / tauE,i` (default 1.0, the common crude $\tau_\phi \approx \tau_{E,i}$ assumption) sets $\tau_\phi = (\text{ratio}) \times \tau_{E,i}$, so treat v_ϕ's absolute scale here as order-of-magnitude, more so than the manual-input mode.

The Dashboard reports v_ϕ (km/s) and a Mach number $M = v_\phi / c_s$ ($c_s = \sqrt{(T_{e0} + T_{i0})/m_i}$, a simple ion-acoustic estimate, not a rigorous sound speed), plus the net torque T_{NBI} for the momentum-balance mode.

1.11 Beam-beam fusion

Off by default (checkbox, Losses section, below the rotation fields). Fast ions from **two separate NBI sources** can fuse with each other directly – a third fusion channel alongside thermal-thermal and beam-target. This is a **reduced** model, not a full treatment: instead of integrating over both beams' slowing-down spectra (a genuine 2-D relative-velocity integral), each population is approximated by a single characteristic velocity at its own average fast-ion energy (`average_fast_energy_keV`, the same representative energy this project already uses for pressure/energy-density diagnostics), signed by its own co/counter direction. Only the **first two** beamlines are paired – this project's tested scope is exactly two NBI sources – summed over every pair of their energy components.

The relevant physical quantity is the beam-beam **relative** velocity:

$$v_{\mathrm{rel}} = |\, s_1 v_1 - s_2 v_2 \,|, \qquad s = +1 \text{ (co)},\ -1 \text{ (counter)},$$

fed into the same deuteron-equivalent-energy cross-section evaluation as the rotation correction to beam-target fusion (Sec. "Bulk toroidal rotation" above) – structurally the same construction, with the *other beam's own velocity* playing the role v_ϕ played there. **This is why counter-injecting the second beam matters physically**: two co-current beams at similar per-amu energy have a small v_{rel} (their velocities nearly cancel) and thus a negligible beam-beam yield, while opposite directions maximize v_{rel} – exactly PONTE's default configuration (D co-current, T counter-current). D-T and D-D pairs are handled (the only cross-section fits this project carries); T-T pairs, or a beam-beam calculation with fewer than two useful beams, contribute nothing.

The result is folded into the existing D-T/D-D fusion power and neutron- rate totals (`pf_dt_w`, `pf_dd_w`, `neutron_rate_s`) exactly like the thermal and beam-target contributions, so it is automatically included in the alpha-heating closure (Sec. "Alpha self-heating" below) with no separate wiring needed. The Dashboard's D-T/D-D fusion-power rows show all three contributions (thermal / beam-target / beam-beam); the Scan tab's "Neutrons" group breaks the neutron rate out the same way, with the beam-beam panel showing a genuine (if usually small, for PONTE's default 0.1 MW counter-beam) nonzero curve once enabled, rather than the flat zero it shows while off.

1.12 Power balance (electron and ion)

At each central density the model solves a 0-D steady-state power balance for T_e and T_i. The heating that drives each temperature is the sum of the neutral-beam split, prescribed auxiliary heating (ECRH and ICRH), and when enabled fusion alpha self-heating.

1.12.1 Beam heating split

Each useful beam j delivers a fraction $L_e(x_j)$ of its power to electrons and $L_i(x_j) = 1 - L_e(x_j)$ to ions, with $x_j = E_{b,j}/E_{c,j}(T_e)$ and

$$L_i(x) = \frac{1}{x}\int_0^x \frac{du}{1+u^{3/2}}, \qquad L_e(x) = 1 - L_i(x).$$

$L_i \to 1$ as $x \to 0$ (cold, ion-dominated) and $L_i \to 0$ as $x \to \infty$ (fast beam, electron-dominated). $P_{\text{useful},j}$ is the injected power after shine-through, first-orbit loss and charge-exchange loss (preceding sections).

1.12.2 Auxiliary heating (ECRH, ICRH)

ECRH and ICRH are entered as powers with electron/ion fractions. Their contribution to each channel is

$$P_{\text{aux},e} = f_e^{\text{EC}} P_{\text{ECRH}} + f_e^{\text{IC}} P_{\text{ICRH}}, \qquad P_{\text{aux},i} = \left(1 - f_e^{\text{EC}}\right) P_{\text{ECRH}} + f_i^{\text{IC}} P_{\text{ICRH}}.$$

ECRH is electron-dominant, f_e^{EC} close to 1. For ICRH the electron and ion fractions are set independently; if $f_e^{\text{IC}} + f_i^{\text{IC}} < 1$ the remainder is treated as power that does not thermalise (a fast-ion tail or direct loss). These sources are Te/Ti-independent constants added to the right-hand side of each balance.

1.12.3 Alpha self-heating (optional)

When enabled, 3.5 MeV fusion alphas deposit

$$P_\alpha = f_\alpha \frac{E_\alpha}{E_f} P_{f,\text{tot}} = f_\alpha \frac{3.5}{17.6} P_{f,\text{tot}},$$

split between electrons and ions by the alpha slowing-down fraction

$$L_e^\alpha = L_e\!\left(\frac{E_\alpha}{E_c^\alpha(T_e)}\right), \qquad E_c^\alpha = \left(\frac{m_\alpha}{m_e}\right)^{1/3} T_e,$$

so $P_{\alpha,e} = L_e^\alpha P_\alpha$ and $P_{\alpha,i} = (1 - L_e^\alpha) P_\alpha$. Because P_α depends on $P_{f,\text{tot}}$, which depends on T_i, which depends on $P_{\alpha,i}$, the alpha term is closed by an under-relaxed fixed-point iteration; with alpha heating off the solve is a single pass.

1.12.4 Decoupled balance (default)

With electron-ion equipartition off, the two balances are solved in sequence. The electron balance is a 1-D root-find for T_e:

$$\frac{3}{2} n_{e0} T_e \, (10^3 e) \, \frac{V}{\tau_{E,e}} = \sum_j L_e(x_j) \, P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e}.$$

The ion temperature then follows in closed form (or, for the neoclassical-ion option, from a second 1-D root-find see *Confinement time*):

$$\frac{3}{2}\, n_{\text{thermal}}\, T_i\, (10^3 e)\, \frac{V}{\tau_{E,i}} = \sum_j L_i(x_j)\, P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i},$$

where $n_{\text{thermal}} = n_{\text{sum}}(n_{e0}, Z_{\text{eff}}) - n_{b0}$ is the thermal-ion density left after the fast ions are subtracted. A non-positive n_{thermal} marks the operating point infeasible.

1.12.5 Coupled balance (equipartition on)

When electron-ion equipartition is enabled, the exchange power $P_{ei}(T_e, T_i)$ (defined in *Electron-ion exchange time*) is subtracted from the electron balance and added to the ion balance, and the pair is solved together:

$$\frac{3}{2} n_{e0} T_e (10^3 e) \frac{V}{\tau_{E,e}} = \sum_j L_e(x_j) P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e} - P_{ei}(T_e, T_i),$$

$$\frac{3}{2} n_{\text{thermal}} T_i (10^3 e) \frac{V}{\tau_{E,i}} = \sum_j L_i(x_j) P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i} + P_{ei}(T_e, T_i).$$

The reported P_e and P_i remain the pure beam split; P_{ei}, P_α and P_{aux} are reported separately.

1.12.6 Profile-corrected 0-D

The balance above is genuinely 0-D: one density, one T_e, one T_i, treated as uniform over the volume. With the **profile-corrected 0-D** checkbox (Plasma section) *off*, the entered central density is used directly as that uniform value and the solved T is a flat-plasma effective temperature; comparing it to a measured on-axis T_0 then needs a peaking factor supplied by the reader (validation runs use $\simeq 2$). In this mode every radial integral (shine-through column density, thermal and beam-target fusion, pressure, thermal energy) also uses **flat** profiles, consistent with the balance: the peaking parameters are ignored. (Before October 2026 these integrals still applied the $(1-\rho^2)^{2p}$ shapes and treated the flat values as on-axis peaks, which with $p_{Te} = 1$ cut the profile-average T to a third and under-reported fusion power by up to $\sim 3\times$.)

With the checkbox *on*, the entered `central n_e` is taken to be the **on-axis** n_{e0}, and for a $(1-\rho^2)^{2p}$ profile the volume average is $\langle X \rangle = X_0/(1+2p)$. The energy balance and the τ_E scalings then run on

$$\langle n_e \rangle = \frac{n_{e0}}{1 + 2\, p_n},$$

so the solved T_e, T_i are **volume-averaged**. The electron and ion temperature profiles are shaped by **independent** exponents p_{Te} (`Temp peaking (electron)`) and p_{Ti} (`Temp peaking (ion)`; a value < 0 means "use p_{Te}"), so the on-axis values are

$$T_{e0} = \langle T_e \rangle\, (1 + 2\, p_{Te}), \qquad T_{i0} = \langle T_i \rangle\, (1 + 2\, p_{Ti}).$$

They are reported (Dashboard row "on-axis T_{e0}/T_{i0}") and are what the fusion, pressure and β_t integrals use as their central values the reactivity integrals with the $(1-\rho^2)^{2p_{Ti}}$ ion shape, the beam slowing-down kernel with the $(1-\rho^2)^{2p_{Te}}$ electron shape. Set p_n (`Density peaking`), p_{Te} and p_{Ti} to match the real profiles; $p \simeq 0.5$ to 0.75 gives $X_0/\langle X \rangle \simeq 2$ to 2.5 (L-/H-mode), while a **hot-ion mode** on a compact ST needs $p_{Ti} \gg p_{Te}$ ($T_{i0}/\langle T_i \rangle \simeq 3$ to 4 with T_e nearly flat). With all peaking parameters at 0 the correction is a no-op.

1.12.7 Deposition-weighted fast-ion profile

With profile-corrected 0-D on, the fast-ion density is built **locally** from the beam deposition (October 2026) instead of a single 0-D value spread like the slowing-down time:

$$n_b(\rho) = \sum_j S_j \, h_j(\rho) \, \tau_{s,j}\big(n_e(\rho), T_e(\rho)\big), \qquad S_j = \frac{P_{\mathrm{useful},j}}{E_{b,j}\, e},$$

with $\tau_{s,j}$ the local thermalisation time and $h_j(\rho)$ the confined birth rate per unit volume of beam j, normalised so that $V\langle h_j\rangle = 1$ (`physics.beam_birth_density_profile`):

- **Attenuation** along tangential chords (Beer-Lambert) on the $n_{e0}(1-\rho^2)^{2p_n}$ profile, with the same stopping cross-section as the shine-through model.
- **Beam cross-section** (`BeamParams.beam_width_m`, `beam_height_m`; HOT-Jass: NBI "width W", "height H"). The beam is bounded by its duct aperture, an ellipse of full width W (horizontal) and height H (vertical); $W = H$ is a circular aperture. Inside it the power profile is Gaussian, $p(u,v) \propto \exp[-u^2/2\sigma_u^2 - v^2/2\sigma_v^2]$, truncated at the aperture edge, which sits at 2σ (`BEAM_EDGE_SIGMA`; $\sigma_u = W/4$, $\sigma_v = H/4$): the profile there is still $e^{-2} \approx 14\,\%$ of the peak, and zero outside. All of P_{NB} passes inside the aperture. $W = H = 0$ (default) is a single pencil ray; one of them 0 is a line of rays along the other. Otherwise `physics.beam_ray_bundle` samples the aperture with parallel rays shifted in tangency radius and height: a rectangular grid spanning it, clipped to the ellipse (every ray inside), with Gaussian weights summing to 1 and spacing $\leq 0.1\,a$ (odd, 3 to 21 rays per axis, so the axis ray is always kept). A single tangential ray puts a finite number of ions into the vanishing volume of its tangency surface $\rho(R_t)$, a singular birth density there; the rays have to be closer than the profile smoothing scale (orbit width, ≥ 0.03) for their singular profiles to add up to a smooth one. All rays are evaluated in one vectorised pass (`physics.tangential_chords`), a few ms per beam. The shine-through fraction is averaged over the same rays (`captured_power_fraction(..., width_m, height_m)`; rays that miss the plasma shine through). First-orbit loss keeps the beam-axis chord. Saves from before October 2026 carry a circular 1/e diameter D ($\sigma = D/2\sqrt{2}$); HOT-Jass loads it as the aperture with the same σ, $W = H = \sqrt{2}\,D$.
- **Prompt loss** (already counted in the orbit-loss power), two options (`orbit_loss_deposition`; HOT-Jass always uses `"consistent"`, no switch):
- `"cutoff"` (HI-Jass default, historical): every birth beyond `orbit_cutoff_rho` is removed. That radius is set by the worst-case (trapped) orbit, so at low current, where the banana width approaches the minor radius, it removes far more births than the loss formula removes power (TCV at 150 kA: 72 % of births against 13 % of power), and the confined beam is renormalised onto the remaining core births.
- `"consistent"` (ST orbit models): each birth is weighted by $1 - P_{lost}(\rho, |\lambda_0|)$, the per-birth rule of the pitch-resolved loss formula (`physics.st_orbit_loss_probability`, also used for the lost power), with $|\lambda_0| = |\xi_0|$ (the full-field birth pitch, "Field directions and the beam birth pitch") along each chord. It changes NBCD by about 30 % on TCV, about 6 % on MAST and 6-19 % on NSTX-U. HOT-Jass uses it for both width-based ST models (the P_ϕ model is always consistent; the large-aspect model keeps the cutoff): a cutoff that removes far more births than power is lost is not physical, and the consistent profile agrees better with ASTRA (TCV) and TRANSP (MAST, NSTX-U).
- **Orbit-width smoothing.** Births are spread over half the passing orbit width (at least 0.03 in ρ), as a *volume-weighted* average of the birth density, so that births do not pile up on the axis.

The volume average of $n_b(\rho)$ replaces the 0-D n_{b0} inside the balance, so dilution, the ther-

mal density $n_{th}(\rho) = n_\Sigma(\rho) - n_b(\rho)$ (locally depleted where the fast ions sit) and the thermal and beam-target fusion integrals all use the same fast ions. The solver stores the radial profiles on `OperatingPoint` (`rho_profile`, `nb_profile_m3`, `n_fuel_profile_m3`, `ne_profile_m3`, `Te_profile_keV`, `Ti_profile_keV`, the fusion power densities, and per beam the birth rates $S_j h_j(\rho)$ and birth pitch $\xi_{0,j}(\rho)$). Their volume integrals are exactly the reported totals. With profile-corrected 0-D off the flat model is kept unchanged.

On the ASCOT5/BBNBI PONTE reference case (4 MW D at 80 keV) the deposition model gives 1.24 MW of fusion power against ASCOT's 1.4 MW; the earlier single-n_{b0} model gave 0.87 MW.

1.13 Thermal ion mix (H, D, T)

The thermal ions are a mix of protium, deuterium and tritium with density fractions $f_H + f_D + f_T = 1$ (`PlasmaParams.hydrogen_fraction`, `deuterium_fraction`, `tritium_fraction`; HOT-Jass: PLASMA "H/D/T fraction", checked at Start). Hydrogen does not fuse, but its mass enters everywhere the ion mass does:

- electron-ion energy exchange (a separate H channel next to D and T);
- the neoclassical $\tau_{E,i}$ (effective mass and ion-ion collision frequency);
- the isotope mass of IPB98(y,2), $M_{\text{eff}} = f_H \cdot 1 + f_D \cdot 2 + f_T \cdot 3$, so a pure H plasma has τ_E lower by $(1/2)^{0.19} = 0.88$ than a pure D one;
- the rotation mass density and the NBCD stopping terms.

$f_H = 0$ (default) reproduces the D-T-only results exactly. A plasma with neither D nor T (e.g. the HOT-Jass T-15MD preset: H beams into an H plasma) makes no fusion power; HOT-Jass then shows "no fusion fuel" instead of the fusion plots.

1.14 Particle balance and fuelling

The power balance fixes T_e and T_i at a given density, but says nothing about how that density is sustained. The HOT-Jass web app (**Balance** tab) adds a steady-state **particle balance** for the thermal fuel ions as a post-processing step: it uses the solved operating point and does not feed back into the solve.

1.14.1 Delayed orbit loss by pitch-angle scattering (optional)

`delayed_loss_factor` k (HOT-Jass: MODELS, "Delayed orbit loss k (pitch scattering)"; default 0 = off). The prompt first-orbit loss uses the birth pitch. While a fast ion slows down, pitch-angle scattering, strong below E_c, moves it to other pitches. It is lost only if the new pitch puts it on an unconfined orbit at its radius and current energy: in the core, where orbits are narrow, nothing is lost.

- Pitch memory (the $l = 1$ Legendre decay of the Cordey slowing-down distribution, the same factor as in the NBCD model), $u = v/v_b$:

$$G(u) = \left[\frac{u^3(1+u_c^3)}{u^3+u_c^3}\right]^{\beta/3}, \qquad \beta = Z_{eff}/\bar{Z}$$

  (`physics.pitch_scattering_beta`, thermal ions and the carbon-like impurity only).

- Loss probability after scattering, $\langle P_{lost}\rangle_\xi(\rho, E)$ (`physics.isotropic_orbit_loss_probability`): each orbit model's own per-birth rule averaged over a uniform pitch $\xi \in [-1, 1]$ – the ST orbit widths at energy E, the P_ϕ orbit boundary on 8 points of the flux surface, or the large-aspect $q_*\rho_{Li}$ shift (counter half only). Evaluated at $u = 1$, 0.8, 0.6, 0.45, 0.3, 0.15 and interpolated.
- Survival (cumulative: a lost ion stays lost):

$$P_{surv}(\rho, u) = 1 - \max_{u' \geq u} \min\{1,\ k\,[1 - G(u')]\,\langle P_{lost}\rangle_\xi(\rho, E_b u'^2)\}$$

  $k = 1$ is orbit physics only; $k > 1$ stands in for TF-ripple or MHD (fishbone, Alfven-eigenmode) losses, which are not computed.
- `physics.delayed_loss_survival` samples P_{surv} uniformly in slowing-down time ($y = \ln(u^3 + u_c^3)$, $dt = \tau_{se}\, dy/3$), so τ_{eff}/τ_S is the time average of P_{surv}.

Consequences: τ_{eff} replaces τ_S in $n_b(\rho)$ (dilution, pressure, β); the slowing-down spectrum (beam-target and beam-beam fusion, mean fast-ion energy) and the NBCD integrand J are weighted by P_{surv}; the energy a lost ion still carries, split by $L_i(E/E_c)$ into the electron and ion shares it would have heated, is a separate loss (`OperatingPoint.P_delayed_loss_w`), so $P_{useful} = P_e + P_i + P_{delayed}$; lost ions do not fuel the thermal plasma (particle balance, NBI_OUT SNIBM). No radial fast-ion transport. With `profile_averaging` off (flat 0-D) the same survival is computed on the deposition at uniform n_e, T_e and birth-averaged: τ_{eff}/τ_S scales the 0-D n_b, the energy loss reduces the 0-D P_e, P_i, and the birth-averaged $P_{surv}(u)$ weights the 0-D beam-target spectrum. A uniform T_e is hotter than the real edge, so E_c there is higher, scattering starts earlier and the flat-mode loss is larger than the profile one.

On the presets (2026-10-20, $k = 1$): JET and ITER below 0.05 % of P_{useful}; TCV, T-15MD 0.2-0.4 %; ST40, MAST-U, NSTX-U, PONTE 1-3 % of the power and 5-12 % of the fast ions (lost late, at low energy), I_{NB} -1 to -3 %, T_e about -1 %.

1.14.2 Fast-ion dilution

At fixed n_e, quasi-neutrality makes every fast beam ion displace a thermal ion ($n_{\text{thermal}} = n_{\text{sum}} - n_{b0}$, see *Power balance*). The share of fast ions,

$$f_{\text{dil}} = \frac{n_{b0}}{n_{\text{thermal}} + n_{b0}},$$

reduces both thermal and beam-target fusion, which scale with the thermal fuel densities. It is largest for low-energy beams into a moderate-density, hot plasma (long slowing-down time), e.g. $\sim 14\%$ for the PONTE design point.

1.14.3 Steady-state balance per species

For each thermal species $s = \text{H}, \text{D}, \text{T}$ present, holding the input H:D:T mix at the solved density,

$$S_{\text{ext},s} + S_{\text{beam}\to s} - R_{\text{burn},s} = \frac{N_s}{\tau_p^*}, \qquad N_s = \langle n_s\rangle V, \qquad \tau_p^* = \left(\frac{\tau_p^*}{\tau_E}\right)\tau_{E,e},$$

where

- N_s is the thermal inventory of species s (volume-average density from the balance times the plasma volume), and N_s/τ_p^* its particle loss rate;

- τ_p^* is the **effective** particle confinement time including recycling, entered as the ratio τ_p^*/τ_E (Plasma input, default 2). In a low-recycling regime $\tau_p^* \approx \tau_p$; with strong recycling $\tau_p^* \gg \tau_p$;
- $S_{\text{beam}\to s}$ is the rate at which beam ions thermalise into their own species (a D beam feeds D): $S_{\text{beam}} = \sum_j P_{\text{useful},j}/(E_{b,j}\, e)$, minus the fast ions burnt in beam-target reactions before they thermalise;
- $R_{\text{burn},s}$ is the thermal ions consumed by fusion: one D and one T per D-T reaction (thermal-thermal reactions consume one of each thermal species; beam-target reactions consume the fast ion and one thermal ion of the other species). H never burns, and D-D burn-up ($\sim 10^{-3}$ of D-T) is neglected;
- $S_{\text{ext},s}$ is the external fuelling (gas puff or pellets) needed to hold the mix. It is also given as a gas throughput, using 1 Pa m^3 of D_2/T_2 at 273 K $= 5.3 \times 10^{20}$ atoms.

1.14.4 Over-supplied species

$S_{\text{ext},s} < 0$ means the beam alone delivers more of species s than the plasma loses: the mix can then only be held by pumping. With no external fuelling of that species its thermal fraction settles at

$$f_s = \frac{(S_{\text{beam}\to s} - R_{\text{burn},s})\,\tau_p^*}{\sum_{s'} N_{s'}}.$$

This is the typical situation for D beams into a tritium-rich target (PONTE: $S_{\text{beam}}(\text{D}) = 4.85 \times 10^{20}$ s^{-1} against a D loss of 2.85×10^{20} s^{-1} at D:T = 10:90, so the D fraction would rise to $\simeq 0.17$ with tritium-only fuelling). A beam species that is not in the mix at all (fraction 0, e.g. an H beam into a D-T plasma) has to be pumped entirely.

1.14.5 Power balance view

The same tab shows the power balance of *Power balance (electron and ion)* as two pies: heating sources ($P_{\text{NB},e}$, $P_{\text{NB},i}$, $P_{\text{aux},e}$, $P_{\text{aux},i}$, P_α) and transport losses $W_e/\tau_{E,e}$, $W_i/\tau_{E,i}$, with $W_e = \frac{3}{2}\langle n_e\rangle\langle T_e\rangle V$ and $W_i = \frac{3}{2} n_{\text{thermal}}\langle T_i\rangle V$. In steady state the two losses sum to the total heating; P_{ei} is an internal exchange, not a loss.

1.14.6 Limitations

- The result scales directly with τ_p^*, which is an input, not computed; the absolute fuelling rates are only as good as that choice. The sign of $S_{\text{ext},s}$ (whether the beam over-supplies a species) is far less sensitive.
- D and T are assumed to share one τ_p^*; isotope effects on particle transport and wall recycling are not modelled.
- Volume-averaged (0-D) balance: no fuelling-depth or density-profile effects (pellet versus gas puff).

1.15 Non-inductive current (NBCD and bootstrap)

The HOT-Jass web app (**Current** tab) estimates the neutral-beam driven and the bootstrap current of a solved operating point (`hotjass.current`, `HotJassModel.current_drive(op)`). It is a **diagnostic only**: nothing feeds back into the solve, so an operating point whose non-inductive current exceeds I_p is flagged rather than corrected. With profile-corrected 0-D it uses the solver's radial profiles, the beam deposition and the birth pitch. With it off (flat 0-D) the solver

still computes the deposition and birth pitch at uniform n_e (`OperatingPoint.deposition_rho`; `rho_profile` stays empty): the NBCD is then evaluated at the uniform n_e, T_e of the 0-D solve, and the bootstrap current is exactly zero (flat profiles have no gradients). The flat NBCD differs from the profile one (colder core, lower E_c, different τ_s), e.g. MAST-U 75 vs 131 kA, JET 1214 vs 1187 kA.

The **Beam** tab shows, next to each beam's birth pitch $\xi_0(\rho)$, the mean pitch of the slowed-down fast ions: the same $l = 1$ pitch memory averaged over the steady slowing-down population (number-weighted), with the delayed-orbit-loss survival when that is on,

$$\langle\xi\rangle = \xi_0 \frac{\int_0^1 P_{surv}\, G^{\beta/3}\, u^2 du/(u^3+u_c^3)}{\int_0^1 P_{surv}\, u^2 du/(u^3+u_c^3)}, \qquad G = \frac{u^3(1+u_c^3)}{u^3+u_c^3}.$$

On the presets $\langle\xi\rangle/\xi_0 \approx 0.5$-$0.85$, lowest in the hot core (higher E_c: more of the slowing-down is scattering-dominated).

1.15.1 Neutral-beam current drive

The fast-ion current of each beam is the local 1-D Cordey / Start-Cordey result for a mono-energetic source, with electron and ion drag and ion pitch-angle scattering (no energy diffusion):

$$j_f = Z_b\, e\, S\, h(\rho)\, \tau_s\, v_b\, \xi_0\, J(u_c, \beta), \qquad J = \int_0^1 \frac{u^3}{u^3+u_c^3} \left[\frac{u^3(1+u_c^3)}{u^3+u_c^3}\right]^{\beta/3} du,$$

where:

- $S\,h(\rho)$ is the confined birth rate per volume (above), and τ_s the Spitzer slowing-down time ($dv/dt = -(v/\tau_s)(1+v_c^3/v^3)$).
- ξ_0 is the full-field birth pitch ("Field directions and the beam birth pitch"; $\approx R_t/R$ for a horizontal midplane chord with weak B_p); it is averaged over the same sub-chords as h and weighted by births, and is negative for counter-current beams.
- $u = v/v_b$, $u_c^3 = (E_c/E_b)^{3/2}$, with the critical energy from the local composition (D, T and the carbon-like impurity), $E_c = 14.8\, A_b T_e \left(\sum_j n_j Z_j^2/n_e A_j\right)^{2/3}$.
- $\beta = Z_{\text{eff}}/\bar{Z}$, with $\bar{Z} = \sum_j n_j Z_j^2 (m_b/m_j)/n_e$, is the ratio of the ion pitch-angle scattering rate to the ion drag on the speed. (Written as $l(l+1)\beta'/3$ with $\beta' = Z_{\text{eff}}/2\bar{Z}$ elsewhere in the literature; for the $l = 1$ moment both give the exponent $Z_{\text{eff}}/3\bar{Z}$.)

The electron shielding with the trapped-electron correction (Start & Cordey 1980) gives the net driven current:

$$j_{NB} = j_f \left[1 - \frac{Z_b}{Z_{\text{eff}}}(1-G)\right], \qquad G = \left(1.55 + \frac{0.85}{Z_{\text{eff}}}\right)\sqrt{\epsilon} - \left(0.2 + \frac{1.55}{Z_{\text{eff}}}\right)\epsilon,$$

with $\epsilon = \rho a/R_0$ and G clipped to $[0, 1]$. Total currents and the current-drive efficiency:

$$I = \frac{V}{2\pi R_0}\langle j\rangle, \qquad \eta_{CD} = \frac{I_{NB}\, R_0\, \bar{n}_{e,20}}{P_{NB}} \quad [10^{20}\,\text{A}\,\text{W}^{-1}\text{m}^{-2}],$$

with P_{NB} the injected power and $\bar{n}_e$ the line-average density.

1.15.2 Bootstrap current

Sauter, Angioni & Lin-Liu, *Phys. Plasmas* **6** (1999) 2834 (erratum **9** (2002) 5140), valid at arbitrary aspect ratio and collisionality, in the cylindrical form

$$j_{BS} = -\frac{1}{B_p}\left[L_{31}\frac{dp}{dr} + L_{32}\, n_e \frac{dT_e}{dr} + L_{34}\,\alpha\, n_i \frac{dT_i}{dr}\right],$$

with p the **thermal** pressure (fast ions excluded; their current is the NBCD), $L_{31}, L_{32}, L_{34}, \alpha$ the Sauter fits in the trapped fraction f_t (Lin-Liu & Miller), ν_e^*, ν_i^* and Z_{eff}, and $dr = a\, d\rho$. The profile gradients are taken analytically from the $(1-\rho^2)^{2p}$ shapes and the current is integrated on a fine grid clustered at the edge: for low peaking exponents ($2p < 1$) the gradients diverge at $\rho = 1$, and a finite-difference gradient would make the edge contribution grid-dependent.

The 0-D model has no current profile, so B_p and q come from an **assumed** one:

$$j \propto (1-\rho^2)^{\nu}, \qquad \nu = \frac{q_{cyl}}{q_0} - 1, \qquad B_p = \frac{\mu_0 I(\rho)}{L_p}, \quad L_p = 2\pi\rho a\sqrt{\frac{1+\kappa^2}{2}},$$

with $I(\rho) = I_p[1-(1-\rho^2)^{\nu+1}]$, q_{cyl} the cylindrical edge q (`safety_factor_cyl_edge`) and q_0 an input (Plasma row `q0`, default 1). The low-aspect-ratio toroidal enhancement of q (q_{95}/q_{cyl}, large on an ST) is not a current-peaking effect, so it enters $q(\rho)$ only, used in the collisionalities:

$$q(\rho) = q_{cyl}(\rho)\left[1 + \left(\frac{q_{95}}{q_{cyl}} - 1\right)\rho^2\right].$$

(Taking ν from q_{95} instead gave $\nu \approx 16$ on MAST-U and NSTX-U, an unrealistically peaked current.)

The inductive remainder is $I_{ind} = I_p - I_{NB} - I_{BS}$ (negative: over-drive), and $f_{NI} = (I_{NB} + I_{BS})/I_p$.

1.15.3 Validation

TRANSP, ASTRA and CRONOS references (details, figures and scripts in `docs/Validation/HI-Jass_validation.md`, sec. F-H). NSTX-U, MAST and TCV use the centre-of-mass cross-sections and are given for both deposition options (cutoff / consistent), with τ_E tuned to the reference stored energy or temperatures; ITER and JET at matched β_N (cutoff, earlier runs):

Case	I_{NB} model / ref.	I_{BS} model / ref.
NSTX-U 0.87 MA, 10.2 MW 80 kV, H98=1	353 / 285 / 287 kA	594 / 621 / 583 kA
NSTX-U 0.975 MA, 12.6 MW 90 kV, H98=1	404 / 327 / 370 kA	653 / 682 / 604 kA
NSTX-U 1.1 MA, 15.6 MW 100 kV, H98=1	466 / 381 / 396 kA	762 / 793 / 704 kA
NSTX-U 0.635 MA, 0.75 T, 6.8 MW 80 kV	199 / 211 / 184 kA	441 / 444 / 451 kA
MAST #18808 0.62 MA, 3.5 MW off-axis	287 / 268 / 253 kA	147-161 / ≈ 45 kA
TCV #59429 0.15 MA, 1.15 MW (ASTRA)	91-124 / 67-89 / ≈ 49 kA	74-80 / ≈ 81 kA
ITER hybrid 12 MA, β_N 2.65	3.33 / 3.5 MA	5.82 / 4.7 MA
ITER steady state (no ITB) 10 MA, β_N 2.70	4.31 / 3.6 MA	5.45 / 5.0 MA
JET #58323 1.4 MA, β_N 2.8	0.55 / 0.33 MA	0.51 / 0.32 MA

References: NSTX-U, Gerhardt et al., *Nucl. Fusion* **52** (2012) 083020 (fully non-inductive TRANSP scenarios, Table 2, broad profiles; W_{tot} matched, f_{NI} = 1.03-1.07 with the consistent deposition). MAST, Turnyanskiy et al., *Nucl. Fusion* **49** (2009) 065002 (TRANSP classical; the bootstrap value is read off a stacked plot). TCV, Piron et al., *Nucl. Fusion* **59** (2019) 096012

(ASTRA profiles, digitised and integrated, about 20 % uncertainty; ECCD not modelled). ITER, Parail et al., *Nucl. Fusion* **53** (2013) 113002 (Table 3). JET, Sips, EPS 2003 O-1.3A (TRANSP; density and beam set-up only approximately known, $a = 0.95$ m used for the diverted plasma).

With the consistent deposition NBCD agrees within about 15 % on NSTX-U and 6 % on MAST; TCV stays 1.4-1.8x high (charge-exchange loss the main uncertainty). The cutoff deposition over-predicts NBCD wherever the orbits are wide. The bootstrap current agrees within about 15 % on NSTX-U and TCV and 25 % at ITER, but is about 3x the MAST reference and high on JET. Matching β_N instead of the stored energy raises both NSTX-U currents by 20-30 % (the codes normalise β_N differently), so cross-code comparisons should match W. Remaining systematic effects: the **full-energy-only beam** ($E/2$ and $E/3$ not modelled; on NSTX-U they lowered I_{NB} by about 15 % in an earlier test) and a first-orbit loss of 25-40 % on spherical tokamaks, well above TRANSP's few per cent.

1.15.4 Limitations

- Each energy component (E, $E/2$, $E/3$) is a separate mono-energetic source; no anomalous fast-ion diffusion, no current diffusion (steady state assumed).
- Assumed current profile and $q(\rho)$; no equilibrium.
- Where fast ions make the thermal-ion density hollow (fixed n_e, large n_b in the core), the thermal bootstrap reverses near the axis (seen on PONTE).
- Low-peaking presets ($p \simeq 0.1$ to 0.25) put roughly 28 % of I_{BS} at $\rho > 0.95$: the power-law shapes act like a pedestal.

1.16 Confinement time

$\tau_{E,e}$ and $\tau_{E,i}$ in the balances above are set per channel by the *Confinement* selector in the Losses section; the electron and ion channels are chosen independently. When a physics-based scaling is selected the manual `tauE,e` / `tauE,i` inputs are ignored, and the Summary sheet reports the computed value (or its range across a scan) instead.

The physics-based scalings below all use the same loss power

$$P_{\text{loss}} = \sum_j P_{\text{useful},j} + P_{\text{aux},e} + P_{\text{aux},i},$$

i.e. the captured, post-orbit, post-charge-exchange beam power plus the prescribed auxiliary (ECRH, ICRH) heating. Fusion alpha heating is not included in P_{loss} (it is closed by an outer fixed-point and is not known when τ_E is evaluated). None of these depend on T_e or T_i, so each is evaluated once.

1.16.1 Confinement enhancement

Every scaling below can be multiplied by a confinement enhancement h (default 0; the MODELS slider under Confinement in the HOT-Jass web app, and the config field `tauE_enhancement` in HI-Jass):

$$\tau_E = (1 + h)\,\tau_{\text{scaling}}, \qquad H = 1 + h.$$

$h = 0$ is the plain scaling. Fixed τ_E input and the neoclassical ion channel are not affected.

1.16.2 Fixed (input)

$\tau_{E,e}$ and $\tau_{E,i}$ are the values entered in the Losses section (under the Confinement selector), used unchanged.

1.16.3 IPB98(y,2) ELMy H-mode

The ITER Physics Basis ELMy H-mode scaling (ITER Physics Basis, *Nucl. Fusion* **39** (1999) 2175; standard reference form):

$$\tau_E = 0.0562\, I_p^{0.93}\, B_t^{0.15}\, n_{19}^{0.41}\, P_{\mathrm{loss}}^{-0.69}\, R_0^{1.97}\, \kappa_a^{0.78}\, \varepsilon^{0.58}\, M_{\mathrm{eff}}^{0.19}$$

with I_p in MA, B_t in T, P_{loss} in MW, R_0 and a in m, $n_{19} = n_{e0}/10^{19}\,\mathrm{m}^{-3}$, $\varepsilon = a/R_0$, $\kappa_a = \kappa$, and $M_{\mathrm{eff}} = 2x_D + 3x_T$ the mass-weighted ion mass number. Selecting *IPB98(y,2) ELMy H-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$) – like every scaling offered here, it does not split electron/ion channels.

The fit was obtained from conventional-aspect-ratio devices ($A \sim$ 2.5-4) and is documented in the spherical-tokamak literature to mis-predict at low aspect ratio. When $A = R_0/a < 2$ the Assumptions tab flags that IPB98(y,2) is being extrapolated outside its dataset.

1.16.4 Kaye NSTX L-mode

An NSTX low-aspect-ratio (spherical tokamak) L-mode global fit (Kaye, *Nucl. Fusion* **46** (2006) 848, Eq. 5 / Fig. 9):

$$\tau_E = 4.73 \times 10^{-4}\, I_p^{1.01}\, B_t^{0.70}\, n_e^{-0.07}\, P_{\mathrm{loss}}^{-0.37}$$

with I_p in A, B_t in T, n_e in m^{-3}, P_{loss} in W and τ_E in s. Selecting *Kaye NSTX L-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$).

1.16.5 Kaye NSTX H-mode

The H-mode counterpart of the previous fit: an NSTX low-aspect-ratio (spherical tokamak) H-mode thermal-energy scaling (Kaye, *Nucl. Fusion* **46** (2006) 848, Table 1, ordinary-least-squares "Case 1" fit to all 85 H-mode points, RMSE = 0.145):

$$\tau_E = 4.69 \times 10^{-9}\, I_p^{0.57}\, B_t^{1.08}\, n_e^{0.44}\, P_{\mathrm{loss}}^{-0.73}$$

with I_p in A, B_t in T, n_e in m^{-3}, P_{loss} in W and τ_E in s. Relative to IPB98(y,2) the current dependence is much weaker ($I_p^{0.57}$ vs $I_p^{0.93}$) and the toroidal-field dependence much stronger ($B_t^{1.08}$ vs $B_t^{0.15}$), the qualitative spherical-tokamak trend; the density and heating-power exponents ($n_e^{0.44}$, $P_{\mathrm{loss}}^{-0.73}$) are of the same order as the conventional-aspect scalings. This is an ST-appropriate fit ($A \sim$ 1.3 to 1.5 dataset). Selecting *Kaye NSTX H-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$).

1.16.6 Neoclassical ion

Not currently selectable from the Confinement dropdown – v1 offers only the four modes above (Fixed, Kaye L, Kaye H, IPB98(y,2)), each applied to both channels. The neoclassical estimate below is implemented in `hotjass/solve.py` and was previously reachable via an "X e

/ neoclassical i" hybrid selector entry; that combination is out of scope for v1 and may return in a later version. Documented here for completeness:

A banana-regime order-of-magnitude estimate (Wesson, *Tokamaks*; Chang-Hinton low-collisionality limit) for the ion channel only:

$$\chi_{i,\mathrm{neo}} \simeq \frac{q^2\,\rho_i^2\,\nu_{ii}}{\varepsilon^{3/2}}, \qquad \tau_{E,i} = \frac{a^2}{\chi_{i,\mathrm{neo}}} = \frac{a^2\,\varepsilon^{3/2}}{q^2\,\rho_i^2\,\nu_{ii}},$$

with $\varepsilon = a/R_0$, the elongation-corrected cylindrical safety factor

$$q = \frac{5\,a^2 B_t}{R_0\,I_p[\mathrm{MA}]}\,\frac{1+\kappa^2}{2},$$

the thermal-ion Larmor radius

$$\rho_i = \frac{m_{\mathrm{eff}} v_{th,i}}{eB_t}, \qquad v_{th,i} = \sqrt{\frac{2T_i(10^3 e)}{m_{\mathrm{eff}}}}, \qquad m_{\mathrm{eff}} = x_D m_D + x_T m_T,$$

and the ion-ion collision frequency summed over D and T,

$$\nu_{ii} = \sum_{s\in\{D,T\}} 4.80\times10^{-8}\,\frac{n_s[\mathrm{cm}^{-3}]\,\ln\Lambda}{\sqrt{\mu_s}\,T_i[\mathrm{eV}]^{3/2}}, \qquad \mu_s = m_s/m_p.$$

Since $\tau_{E,i} \propto T_i^{1/2}$, the ion balance becomes a 1-D fixed-point root-find for T_i rather than a closed-form substitution. Pure neoclassical transport has no anomalous channel, so it under-estimates real ion transport and can predict very hot, sometimes implausible T_i; treat it as a lower bound. It cannot be combined with electron-ion equipartition.

1.16.7 Neoclassical ion (arbitrary aspect ratio)

Also not currently selectable – same v1 scope note as above.

The bare $\varepsilon^{-3/2}$ geometric factor above both diverges as $\varepsilon \to 0$ and stays bounded as $\varepsilon \to 1$, neither of which is right on a spherical tokamak. The *arbitrary A* variant (previously, selector entries ending "*neoclassical i (arbitrary A)*") replaces the trapped-fraction part with the Lin-Liu & Miller f_t/f_c the banana-regime structure of Helander & Sigmar's *Collisional Transport in Magnetized Plasmas*, Ch. 11 and uses the arbitrary-A edge safety factor q_a:

$$\chi_{i,\mathrm{neo}} \simeq q_a^2\,\rho_i^2\,\nu_{ii}\,\frac{f_t}{f_c}, \qquad f_t = 1 - \frac{(1-\varepsilon)^2}{(1+1.46\sqrt{\varepsilon})\sqrt{1-\varepsilon^2}}, \qquad f_c = 1 - f_t,$$

$$q_a = \frac{5\,a^2 B_t}{R_0\,I_p[\mathrm{MA}]}\,\frac{1+\kappa^2(1+2\delta^2-1.2\delta^3)}{2}\,\frac{1.17-0.65\,\varepsilon}{(1-\varepsilon^2)^2}.$$

f_t/f_c stays finite for every $\varepsilon < 1$ and rises steeply only as $\varepsilon \to 1$ (all ions trapped) the physically correct low-aspect trend. This is **not** a smooth reduction of the $\varepsilon^{-3/2}$ form: the two order-of-magnitude estimates can differ by up to about one order of magnitude in either direction (more once the $\tau_{E,i} \propto T_i^{1/2}$ fixed point amplifies it), which is exactly why both are offered run them side by side. References: Helander, *Phys. Plasmas* **7** (2000) 3999; Hinton, Wiley *et al.*, *Phys. Rev. Lett.* **29** (1972) 698; Satake *et al.*, *Phys. Plasmas* **9** (2002); Goldston & Rutherford (1995).

1.16.8 Known limitation: the P fed into IPB98(y,2) / Kaye NSTX

These three scalings use $P_{\text{loss}} = \sum_j P_{\text{useful},j} + P_{\text{aux},e} + P_{\text{aux},i}$ (sec. above) NBI power *after* shine-through, first-orbit loss **and** charge-exchange loss are all subtracted. The experimental databases these scalings were fit to use a coarser convention instead (absorbed power, i.e. shine-through-corrected only they do not subtract orbit-loss/CX-loss of the already-absorbed fast-ion population), so HI-Jass currently under-feeds P into these negative-exponent scalings and is biased toward *overestimating* τ_E whenever orbit-loss/CX-loss are significant (STs especially). Not yet fixed. See **Appendix A** for a quantitative comparison, including how this compares to the (much larger) effect of the scaling choice itself.

1.17 Mean fast-ion energy

Each beam has its own steady-state slowing-down distribution:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \qquad 0 < E \le E_{b,j}$$

Its mean fast-ion energy is:

$$\langle E_{fast} \rangle_j = \frac{\int_0^{E_{b,j}} E f_j(E)\, dE}{\int_0^{E_{b,j}} f_j(E)\, dE}$$

1.18 Combined values shown in Results

The GUI reports power-weighted effective values for the two beams.

Effective slowing-down time:

$$\tau_S^{\text{eff}} = \frac{\sum_j P_{\text{useful},j} \tau_{S,j}}{\sum_j P_{\text{useful},j}}$$

Effective mean fast-ion energy:

$$\langle E_{fast} \rangle = \frac{\sum_j P_{\text{useful},j} \langle E_{fast} \rangle_j}{\sum_j P_{\text{useful},j}}$$

These effective values are used for the combined fast-ion diagnostics, including fast-ion energy density, pressure, beta, and:

$$R = \frac{u_{fast}}{U_t}$$

The underlying HotJass calculation still evaluates each beam separately before forming these combined diagnostics.

1.19 Plasma beta: isotropic and anisotropic fast ions

Toroidal and normalised beta use the volume-averaged pressure of thermal electrons, thermal ions and fast beam ions:

$$\beta_t = \frac{\langle p \rangle}{B_0^2/2\mu_0}, \qquad \beta_N = \beta_t[\%]\,\frac{a\,B_0}{I_p[\mathrm{MA}]}, \qquad \langle p \rangle = \langle n_e T_e \rangle + \langle n_i T_i \rangle + p_{fast}.$$

The fast-ion pressure depends on the pitch distribution, which the 0-D model does not resolve. With $u_{fast} = n_b \langle E_{fast} \rangle$ the fast-ion energy density and $\xi = v_\parallel / v$:

$$p_{fast} = \left(1 - \langle \xi^2 \rangle\right) u_{fast}.$$

- **Isotropic** ($\langle \xi^2 \rangle = 1/3$, the default): $p_{fast} = \frac{2}{3} u_{fast}$.
- **Anisotropic** bound ($\langle \xi^2 \rangle = 1$, all fast ions parallel): $p_{fast} = 0$.

The solver reports the first as `beta_t` (pitch set by `fast_ion_mean_pitch2`) and always computes the second as `beta_t_anisotropic`.

Both are shown (the anisotropic value in brackets). Tangential beams are born nearly parallel and isotropise only by pitch-angle scattering while slowing down, so the true value lies between the two bounds: closer to the anisotropic one for tangential injection at high T_e (slowing down faster than scattering), closer to the isotropic one for perpendicular beams. The difference matters where fast ions carry a large share of the pressure (low density, high beam power). β is a diagnostic: nothing in the solve depends on it.

Fast-ion pressure in the NBI_OUT export (HOT-Jass web). The parallel and perpendicular pressures written to NBI_OUT (PBLON, PBPER) follow the *Fast-ion pressure* choice at the end of MODELS:

$$p_\parallel = 2\langle \xi^2 \rangle\, u_{fast}, \qquad p_\perp = \left(1 - \langle \xi^2 \rangle\right) u_{fast}, \qquad p_\perp + p_\parallel/2 = u_{fast},$$

so the stored energy does not depend on it. *Isotropic* (1/3) and *Anisotropic (parallel)* (1) are the two bounds above. The default, *Birth pitch + scattering*, takes a local $\langle \xi^2 \rangle(\rho)$ per beam component from the steady slowing-down distribution of the NBCD model (electron and ion drag plus ion pitch-angle scattering, see *Neutral-beam current drive*). The $l = 2$ Legendre moment of the pitch starts at the birth value and decays as G^β while the ion slows down ($G^{\beta/3}$ is the $l = 1$ factor of the NBCD integral); weighting by energy over the distribution $\propto u^2/(u^3 + u_c^3)$:

$$\langle \xi^2 \rangle = \frac{1}{3} + \frac{2}{3}\,P_2(\xi_0)\,\frac{\int_0^1 \frac{u^4}{u^3+u_c^3}\,G^\beta\,du}{\int_0^1 \frac{u^4}{u^3+u_c^3}\,du}, \qquad G = \frac{u^3(1+u_c^3)}{u^3+u_c^3}, \qquad P_2 = \frac{3\xi_0^2 - 1}{2},$$

with $u = v/v_b$, u_c, $\beta = Z_{eff}/\bar{Z}$ and the profiles exactly as in the NBCD calculation and $\xi_0(\rho)$ the birth-weighted mean pitch (the pitch spread within a radial bin is neglected). Beam-averaged values on the presets: NSTX-U 0.23 / 0.49 (NBI-1 is born at low pitch, more perpendicular than isotropic), MAST-U 0.50 / 0.57, ITER 0.49 / 0.55 – between the bounds, as expected for tangential beams.

1.20 HOT-Jass web: new machine, minimal input and input checks

The HOT-Jass web app starts in the **NEW MACHINE** state: every input cell is empty and every model is at its simplest option (fixed τ_E, Manual shine-through, large-aspect orbits, manual CX fraction, rotation off, all optional physics off except the profile-corrected 0-D). A step-by-step introduction for new users – seven numbers that define a tokamak, and a machine built from the empty rail – is the **Beginner guide** (`docs/HOT-Jass_Beginner_Guide.pdf` in the HOT-Jass web repository, also offered in its Help dialog).

1.20.1 Minimal input

A cell may be left empty when the calculation does not need it. Each cell is, given the other inputs:

- **required** (yellow while empty; blocks Start): R_0, a, B_0, I_p, n_{e0}; $\tau_{E,e}$ and $\tau_{E,i}$ with Confinement = Fixed; E_b and tangent R of a beam whose power is above 0. At least one heating power (P_{NB}, P_{ECRH}, P_{ICRH}) must be above 0;
- **default** (the empty cell shows its value in gray italics; the defaults used are listed at Start): $\kappa = 1$, $\delta = 0$, $Z_{\mathrm{eff}} = 1$, density and temperature peaking 1, ion peaking -1 (same as electrons), centre-post $R = -1$ $(R_0 - a)$, $q_0 = 1$, $\tau_p^*/\tau_E = 2$, powers 0, beam species D, beam size, Z_t and vertical angle 0, manual shine-through and CX fraction 0, ECRH $f_e = 1$, ICRH $f_e = f_i = 0.5$;
- **not used** (gray): fields of a beam that is off, ECRH/ICRH fractions at zero power, τ_E inputs with a scaling law, the CX and rotation parameters of models not selected, the manual shine-through fraction unless Shine-through = Manual.

Fractions that must add up – H + D + T = 1, each beam's E + E/2 + E/3 = 100 %, ICRH $f_e + f_i = 1$ – take their defaults when the whole group is empty (pure D; mono-energetic; 0.5/0.5). If exactly one cell of a group is empty it takes the remainder; otherwise empty cells count as 0.

1.20.2 Blocking errors

Start is refused, with the reason in the Input check dialog, for an empty required cell, an unparseable or out-of-range value, a group that does not add up, $a \geq R_0$, and a beam aimed outside the plasma ($R_t > R_0 + a$; with the Manual shine-through model such a beam would otherwise still be counted as absorbed).

1.20.3 "Worth checking" flags

Allowed but probably unintended inputs and results are flagged in orange; they never block Start.

Flag	Condition	Where
q_0 has no effect	$q_0 \geq q_{\mathrm{cyl}}$ (flat assumed current profile)	before Start
q_0 below 1	$q_0 < 1$ (sawteeth would hold $q_0 \approx 1$)	before Start
Kink limit	$q_{95} < 2$	before Start
Greenwald	line-average $n_e > n_{\mathrm{GW}}$	before Start
Beam hits the centre-post	$R_t <$ centre-post radius	before Start
Orbit model	$R_0/a < 2$ with the large-aspect model	before Start
Peaking	peaking set with profile-corrected 0-D off	before Start
Shine-through	$> 25\,\%$ of P_{NB}	Power tab
First-orbit loss	$> 30\,\%$ of P_{NB}	Power tab
β_N	> 3.5 (> 5.5 for $R_0/a < 2$)	Plasma tab
Fast ions	$> 10\,\%$ of all ions	Beam tab

The PONTE preset uses a **demanding** profile: $q_{95} < 3$, $n/n_{\mathrm{GW}} > 0.8$, shine-through $> 10\,\%$, first-orbit loss $> 15\,\%$, fast ions $> 5\,\%$, $\beta_N > 3.0$ (4.5 for $R_0/a < 2$), plus checks of the model choices (fixed τ_E, equipartition off, an orbit model other than P_ϕ, D and T beams without beam-beam fusion) and of the result ($T_{i0}/T_{e0} > 2$, $T_{i0} > 30$ keV, thermal ions below 70 %, fast-ion energy density above the thermal one, $f_{NI} > 1$, shine-through above 0.5 MW).

1.21 Results tabs (Operating point)

- **Dashboard** scalar table of the operating point (temperatures, and their on-axis values when profile-corrected 0-D is on; densities, power balance terms, orbit widths, D-T / D-D fusion power, neutron rate, Greenwald ratio, edge safety factor q_*, beta). q_* is reported using whichever formula the active orbit-loss model actually uses – the bare cylindrical `safety_factor_cyl_edge` for "large-aspect", or the arbitrary-A `safety_factor_cyl_edge_arbitrary_A` (q_a) for either ST orbit variant – so it stays consistent with the orbit widths shown just above it. Per-beam CX-loss rows show $f_{\mathrm{cx},P}/f_{\mathrm{cx},N}/\gamma_{\mathrm{cx}}$ (plus n_0 and the informational escape probability) when a physics-based CX model is selected, or just the flat fraction for "Manual fraction" – see "Charge-exchange loss" above. The rotation row shows "off" unless a Rotation model is selected, in which case it shows v_ϕ, a Mach number, and (momentum-balance mode) the net NBI torque – see "Bulk toroidal rotation" above.
- **Deposition** (1) beam targeting geometry in the torus top view (titled with the active device), tangent to each beam's R_t with its co-/counter-current sense; (2) neutral-beam survival $I(s)/I_0$ and the fast-ion birth rate $n_e\sigma\, I(s)/I_0$ along the beam, annotated with each beam's shine-through percentage; (3) the resulting fast-ion birth versus normalised radius ρ, per beam and total (chord-sampled, midplane circular approximation), with the prompt first-orbit-loss zone $\rho > \rho_{cut}$ shaded and a per-beam ρ_{cut} line (from the same criteria as the orbit-loss model in use); (4) the steady-state slowing-down distribution $f(E)$ with the E_b edge marked.
- **Power flow** waterfall / Sankey / pie of injected power to its sinks.
- **Profiles** $n_e(\rho)$, $T_{e,i}(\rho)$, $\tau_S(\rho)$ (local thermalization time, from the same local $n_e(\rho)$, $T_e(\rho)$ used elsewhere in this panel – log scale, since it rises sharply toward the edge as $n_e \to 0$), $n_0(\rho)$ (background-neutral density behind the active CX-loss model – see "Charge-exchange loss" above), the plasma shape, and $P_{fus}(\rho)$ (local D-T + D-D fusion power density, thermal vs. beam-plasma / beam-target, built from the same on-axis densities and $(1-\rho^2)^{2p}$ shapes as the volume-integrated totals on the Dashboard – see "Fusion power" below).
- **Fusion** a pie chart of P_{fus} split into its six channel/reaction combinations (D-T and D-D, each thermal / beam-target / beam-beam), with MW values and percentages; slices under 0.5% of the total are hidden to keep the chart readable (still fully available in the Dashboard's D-T/D-D breakdown rows). Title also shows Q and the full neutron rate Y_n.
- **Assumptions** the full validity / fit-range read-out and any warnings.
- **References** the literature behind the *currently selected* models (beam stopping, confinement scaling, orbit-loss model) and the active machine geometry, each with a link, plus a feedback button.

1.22 Saving and loading a run

Below the results tabs, **Export** has three buttons plus a fourth for the reverse direction:

- **Summary PNG / Summary PDF** a snapshot of the Summary tab (the fixed 3x4 scan-panel grid) as an image.
- **Run record (JSON)** every plasma/beam input, every model toggle, the full solved operating point, and the Summary sheet's scan arrays, as one JSON file (needs a completed run; disabled with a status message otherwise).
- **Load run record...** the reverse of the above: pick a previously saved run-record JSON, and every input field and toggle (confinement, orbit loss, CX-loss model, rotation, beam-beam, equipartition, alpha self-heating, profile-corrected 0-D, both NBI sections) is restored from it, then the point is re-solved automatically. Tolerant of a record written by an older or newer HI-Jass version: a field this version doesn't recognise is ignored, and a field this version expects but the record lacks simply keeps its current value instead of raising an

error. Only the solved *inputs* are restored – the record's own `operating_point /summary` output is historical reference only, not re-applied (the fresh re-solve after loading is what actually reproduces those numbers, so loading a record from a much older HI-Jass version may reproduce a slightly different result if the underlying physics changed meanwhile).

1.22.1 U-files for ASTRA / TRANSP (HOT-Jass web)

In HOT-Jass web, **Results → Save → Save U-files** writes the last calculation as ASCII U-files (the TRANSP / ITPA profile-database format read by ASTRA and other transport codes): one quantity per file, all in one zip named `hot_jass_ufiles_<TOK><shot>.zip`, with a `README.txt` listing every file. The shot number is set in the dialog (*Shot #*); the 4-character tokamak tag comes from the active device name. Files are named `<TOK><shot>_<NAME>.2d` / `.1d`.

The operating point is steady state, so every file carries the same values at two times, $t = 0$ and 100 s: a code reading any time in between gets this point. Radial profiles use 51 points in ρ from 0 to 1, where ρ is the HOT-Jass flux label $\sqrt{((R - R_0)/a)^2 + (Z/\kappa a)^2}$ (normalised minor radius on elliptical surfaces), **not** the toroidal-flux ρ_{tor}.

2-D files, $f(\rho, t)$:

Name	Units	Quantity
NE	M**-3	electron density
TE	EV	electron temperature
TI	EV	ion temperature
ZEFF	–	effective charge (flat)
NFAST	M**-3	NBI fast-ion density
QNBIE	W/M**3	NBI heating of electrons
QNBII	W/M**3	NBI heating of ions
SNBII	M**-3/S	NBI fast-ion source (confined births)
PFUS	W/M**3	fusion power density (thermal + beam-target)
CURNBI	A/M**2	NB-driven current density
CURBS	A/M**2	bootstrap current density
Q	–	safety factor of the assumed current profile

1-D files, $f(t)$: IP [A] and BT [T] (signed, positive counter-clockwise seen from above), PNBI [W] (injected NBI power), RGEO and AMIN [M] (R_0, a), KAPPA, DELTA.

Notes:

- QNBIE, QNBII, SNBII and PFUS need profile-corrected 0-D (PLASMA); without it they are not written, and the README says so. CURNBI, CURBS and Q are written in both modes (flat 0-D: NBCD at uniform n_e, T_e, CURBS = 0).
- QNBIE / QNBII put the deposited NBI power at the birth radius and split it between electrons and ions with each beam component's slowing-down fraction (no radial redistribution during slowing down). The README checks $\int (Q_{NBIE} + Q_{NBII})\, dV$ against the useful NBI power of the power balance and warns if they differ: with the historical cutoff deposition, a beam born entirely outside the orbit-loss cutoff radius has no births although its useful power is counted (the large-aspect orbit model only; the ST orbit models use the consistent deposition).
- CURNBI, CURBS and Q are diagnostics on an assumed current profile (Current tab), not a current evolution.
- Format: label lines with `;-...-` descriptions from column 32 (name 20 characters, units 10), then the X grid, the time grid and f with X varying fastest, 6 numbers per line, 13

characters each (`E13.5`, so a negative value keeps a leading blank and both fixed-width and whitespace readers work), then the `;----END-OF-DATA` trailer and free-text comments.

1.22.2 NBI_OUT for ASTRA (HOT-Jass web)

Results → Save → NBI_OUT writes `nbi_out.dat` in the NBI_OUT v1 format of the file-based NBI coupling to ASTRA (module → ASTRA): line 1 `# NBI_OUT v1`; 15 scalar `KEY value` lines in fixed order; `SOURCES` and one row per beamline (source number, PINJ, PSHINE, PION, PLOSS [W], with the E, $E/2$, $E/3$ components added); `NPROF 22`, a `COLUMNS` line and NA1 rows; `END`. SI units (m^{-3}, W, A, Pa, N); reals `ES24.16`, integers `I10`, no comment lines. **NBI_OUT Preview** plots all 21 columns against RHO with the scalars before saving; both use the same data.

HOT-Jass is not called by an ASTRA stub (no NBI_IN), so TIME = NCALL = 0. The grid is NA1 cell centres, set in the dialog (default 100), $\rho_j = (j-\frac{1}{2})/N_{A1}$ of the HOT-Jass elliptical label, written as RHO $= \rho_j\, a\sqrt{\kappa}$ [m] (approximately the toroidal-flux radius $\sqrt{\Phi/\pi B_0}$; not ASTRA's own equilibrium ρ). Volume integrals use $\int f\, dV = V \sum_j f_j\, 2\rho_j \Delta\rho$.

Profile columns, from the solver's profiles and the confined birth rate $S_k h_k(\rho)$ of each energy component k (normalised to its useful power):

Column	Units	Computed as
RHO	m	$\rho_j a\sqrt{\kappa}$
PEBM, PIBM	W/m^3	$\sum_k L_{e,k} S_k h_k E_k$, $\sum_k L_{i,k} S_k h_k E_k$
PBEAM	W/m^3	PEBM + PIBM
NIBM	m^{-3}	$\sum_k S_k h_k\, \tau_{th,k}(n_e, T_e)$
PBLON, PBPER	Pa	$2\langle\xi^2\rangle u$, $(1-\langle\xi^2\rangle)\, u$
CUFI, CUBM	A/m^2	Current tab: unshielded, shielded
SNEBM	$m^{-3}s^{-1}$	$\sum_k S_k h_k$
SNNBM	$m^{-3}s^{-1}$	zeros (not modelled)
SNIBM1, 2, 3	$m^{-3}s^{-1}$	$S_k h_k$ of the H, D, T components
SCUBM	N/m^3	$\pm \sum_k S_k h_k\, m_b v_b R_t / R_0$
NNBM1, 2, 3	m^{-3}	zeros (not modelled)
STNBDP, SDNBDP1/2, SDNBTP	s^{-1}	$n_b \langle\sigma v\rangle$ per target ion

PEBM and PIBM put the deposited power at the birth radius and split it with each component's slowing-down fractions $L_{e,k}$, $L_{i,k}$; PBEAM counts confined births only. NIBM is the solver's fast-ion density. In PBLON and PBPER, $u = \sum_k n_{b,k} \langle E\rangle_k$ and $\langle\xi^2\rangle$ follows *Fast-ion pressure* (see *Plasma beta*). SCUBM is positive along I_p. The beam-target rates use the local slowing-down spectrum: STNBDP T beam on D; SDNBDP1 D on D, t + p branch; SDNBDP2 D on D, n + ^{3}He branch; SDNBTP D beam on T.

Scalars: PINJ, PSHINE, PION, PLOSS (prompt orbit + CX) from the solver's power balance; PHEAT $= \int(\text{PEBM} + \text{PIBM})\, dV$; WFAST $= \int(\text{PBPER} + \text{PBLON}/2)\, dV$; IBM the NB-driven current of the Current tab; TORQ $= \sum 2P_{useful} R_t / v_b$ (also with rotation off); NRATE the beam-target plus beam-beam neutron rate; NSRC the number of beamlines. STATUS: 0 OK; 1 if $|\text{PION} - \text{PHEAT} - \text{PLOSS}| > 10^{-2}\,\text{PION}$, with the difference in MESSAGE (the historical cutoff deposition, or a coarse grid: below about 50 points the grid sum is off by 1–2 %); −3 without profile-corrected 0-D (all columns but RHO zero).

Checks on the presets: PION = PHEAT + PLOSS within 0.1 %; NIBM within 0.5 % of the solver's fast-ion density; $R_0 \int \text{SCUBM}\, dV = \text{TORQ}$; the beam-target columns times the thermal D, T densities reproduce the solver's beam neutron rate. Approximations: global L_e, L_i per component and no radial redistribution while slowing down; ions lost promptly or by CX are not in the sources; SCUBM uses R_0 for the local R; the SDNBDP1/2 branch assignment follows the

v1 specification, which itself lists it as still to be checked against ASTRA.

1.23 Interface font size (accessibility)

The button next to the Operating point/Scan mode toggle, above the Run button, cycles the whole interface through four zoom levels: 100% → 115% → 130% → 150% → back to 100%. Each click rescales every rail label/entry/button, the Dashboard, and all plot text (titles, axis labels, legends, tick labels) together, redrawing the currently solved point immediately so nothing looks mismatched. The chosen level is written to `~/.hi_jass/settings.json` and restored automatically on the next launch.

1.24 Electron-ion exchange time

The electron-ion exchange time is based on the NRL Plasma Formulary thermal-equilibration rate. For each thermal ion species s (D or T):

$$\bar{\nu}_{ei,s} = 1.8 \times 10^{-19} \frac{\sqrt{m_e m_s}\, Z_e^2 Z_s^2\, n_e \ln\Lambda}{(m_s T_e + m_e T_i)^{3/2}}$$

In this practical-unit expression, masses are in grams, temperatures are in eV, and electron density is in cm^{-3}. The model uses $Z_e = Z_s = 1$ for hydrogenic D and T ions.

For a D-T plasma, the effective exchange time is:

$$_{IE} = \frac{n_D + n_T}{n_D \bar{\nu}_{ei,D} + n_T \bar{\nu}_{ei,T}}$$

For a single thermal ion species, this reduces to:

$$_{IE} = \frac{1}{\bar{\nu}_{ei}}$$

The equivalent energy-over-power form used by the application is:

$$_{IE} = \frac{\frac{3}{2}(n_D + n_T)|T_i - T_e|(10^3 e)V}{|P_{ie}|}$$

where V is the plasma volume and $e = 1.602176634 \times 10^{-19}$ J/eV. The exchange power is:

$$P_{ie} = -P_{ei} = -\frac{3}{2} V (T_e - T_i)\left(n_D \bar{\nu}_{ei,D} + n_T \bar{\nu}_{ei,T}\right)(10^3 e)$$

The absolute values make τ_{IE} positive regardless of whether electrons heat ions ($T_e > T_i$) or ions heat electrons ($T_i > T_e$).

1.25 Fusion power

The total fusion power is the sum of the thermal D-T and beam-target contributions:

$$P_f = P_{f,\mathrm{thermal}} + P_{f,\mathrm{beam}}$$

1.25.1 Thermal D-T fusion

The thermal contribution is calculated from the thermal deuterium and tritium densities and the ion-temperature-dependent Bosch-Hale reactivity:

$$P_{f,\mathrm{thermal}} = n_{D0} n_{T0} \langle \sigma v \rangle_{DT}(T_i) V E_f$$

Here n_{D0} and n_{T0} are the thermal species densities, V is the plasma volume, and:

$$E_f = 17.6\,\mathrm{MeV} = 17.6 \times 10^6 e$$

The Bosch-Hale thermal reactivity is evaluated at T_i, because the relative velocity is that of the thermal ions, not the electrons.

1.25.2 Beam-target fusion

Each useful beam is evaluated separately against the stationary thermal ions of the other D-T species:

$$P_{f,\mathrm{beam}} = \sum_j V E_f \int_0^{E_{b,j}} n_{\mathrm{target},j} f_j(E) [\sigma v]_j(E)\, dE$$

The beam slowing-down distribution is normalized to the fast-ion density of that beam:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \qquad \int_0^{E_{b,j}} f_j(E)\, dE = n_{b0,j}$$

The beam-target reactivity is:

$$[\sigma v]_j(E) = \sigma_{DT}(E) v_j(E), \qquad v_j(E) = \sqrt{\frac{2E}{m_j}}$$

Cross-section energy. The Bosch-Hale cross-section fits take the centre-of-mass energy, while every beam-target and beam-beam call passes the lab energy of a deuteron on a target at rest, $\frac{1}{2} m_D v_{rel}^2$. By default (historical) the fits are evaluated at that lab energy. With `cm_energy_correction` (HOT-Jass: MODELS, "COM energy in beam cross-sections") they use

$$E_{cm} = E\,\frac{m_{target}}{m_D + m_{target}}: \qquad E_{cm} = E/2 \text{ (D-D)}, \qquad E_{cm} \simeq 0.6\,E \text{ (D-T)}.$$

The lab-energy default over-predicts D-D beam-target rates (about 3x at 60 keV; MAST neutron rates 2-3.5x TRANSP instead of 0.5-0.9x) and misplaces the D-T resonance (beam-target D-T too low above about 80 keV, x0.5 at 120 keV). Thermal reactivities are not affected. Off by default until checked on more cases.

For a D-T beam-target reaction $E_f = 17.6$ MeV. For a D beam, $n_{\mathrm{target},j} = n_{T0}$; for a T beam, $n_{\mathrm{target},j} = n_{D0}$. The beam fast-ion density is obtained from the useful beam power:

$$n_{b0,j} = \frac{P_{\mathrm{useful},j} \tau_{S,j}}{E_{b,j}(10^3 e) V}$$

The input shine-through fraction is used to determine captured beam power; the plotted P_{shine} is then the derived shine-through power from the scan. Charge-exchange and other configured losses similarly reduce the useful power before calculating $n_{b0,j}$.

The fusion post-processing always uses the $(1-\rho^2)$ profile weighting: the thermal contribution is

$$P_{f,\text{thermal}} = V E_f \left\langle n_D(\rho) n_T(\rho) \langle \sigma v \rangle_{DT}[T_i(\rho)] \right\rangle_V ,$$

and the beam-target contribution uses the local target density and the local slowing-down distribution evaluated with $T_e(\rho)$. The n_{D0}, n_{T0}, T_e, T_i used here are the **on-axis** values (see *Profile-corrected 0-D*).

1.25.3 D-D fusion

The D-D channel (present whenever the deuterium fraction $x_D > 0$) is evaluated with the Bosch-Hale D-D fits, **branch-resolved**:

branch	products	E_f	neutron
D(d,p)T	T (1.01) + p (3.02 MeV)	4.033 MeV	none
D(d,n)ş He	ş He (0.82) + n (2.45 MeV)	3.269 MeV	2.45 MeV

Thermal:

$$P^{\text{th}}_{f,\text{DD}} = V\left[E_{ddp}\left\langle \tfrac{1}{2} n_D^2 \langle \sigma v \rangle_{ddp}(T_i) \right\rangle_V + E_{ddn}\left\langle \tfrac{1}{2} n_D^2 \langle \sigma v \rangle_{ddn}(T_i) \right\rangle_V\right],$$

with the $\frac{1}{2}$ identical-particle factor and the same $(1-\rho^2)$ profile weighting as the D-T thermal term.

Beam-target (a fast D ion on the thermal-D population **no** $\frac{1}{2}$ factor, the two populations are distinct):

$$P^{\text{beam}}_{f,\text{DD}} = \sum_{j \in \text{D beams}} V \int_0^{E_{b,j}} n_{D0}\, f_j(E)\left[E_{ddp}\,\sigma_{ddp}(E) + E_{ddn}\,\sigma_{ddn}(E)\right] v_j(E)\, dE.$$

The cross-sections take the **lab-frame** deuteron energy (same convention as σ_{DT}). D-D does **not** produce alphas, so the alpha self-heating fixed-point closes on the D-T power only ($P_\alpha = f_\alpha(3.5/17.6)P_{f,DT}$).

1.25.4 Neutron rate

$$Y_n = \frac{P_{f,DT}}{17.6\,\text{MeV}} \;+\; R^{\text{th}}_{ddn} + R^{\text{beam}}_{ddn}\ (+\,Y^{\text{bb}}_n),$$

i.e. one 14.06 MeV neutron per D-T reaction plus the 2.45 MeV neutrons from the D(d,n) branch, plus the beam-beam contribution when that's enabled (see "Beam-beam fusion" above). Named Y_n, not R_n, to avoid confusion with the fast/thermal energy-density ratio R reported elsewhere. Reported on the Dashboard, broken down by channel (beam-target / thermal / beam-beam / full) in the Scan tab's `Y_beam, Y_thermal, Y_bb, Y_full` group and the Summary tab's "Neutron rate" panel, and as a slice of the Fusion tab's pie chart.

1.25.5 Current model limitations

- The beam-target integral uses stationary target ions, so the solved T_i is not included in that reaction integral, and the beam is treated as a single full-energy component (no half/third fractions).
- The tritium and şHe bred by D-D are not burned further ("cat-DD" is neglected).
- The radial volume weighting uses the model's elliptical flux-surface approximation and does not yet solve a 2D equilibrium.

1.26 Density scan

The Plasma tab defines the requested scan range with `n_e_min` and `n_e_max`. The Results tab reports the physically valid interval separately. Points below the HotJass charge-neutrality feasibility limit are not assigned physical output values and are omitted from plots.

The T_e, T_i scan plot's y-axis top is $\min(\max(T_e, T_i), 100\,\mathrm{keV})$ – the actual data maximum, unless it exceeds 100 keV (the upper edge of the Bosch-Hale fits' validated range), in which case the axis is capped there so one runaway hot-ion point doesn't wash out the rest of the curve. The bottom is fixed at 0 (not autoscaled) so a fixed top never stretches the axis into an apparent negative margin.

The n_D, n_T, n_{b0} scan group's otherwise-empty 4th panel shows the fast-ion / target-ion density ratio $n_{b0}/n_{\mathrm{Target}}$, where n_{Target} is whichever thermal D-T species the *dominant* (by injected power) NBI beam reacts with in the beam-target channel – n_T for a D beam, n_D for a T beam.

Four further scan groups:

- P_e, P_i, P_{ie}, Q ("Powers") – $Q = P_{f,tot}/P_{NB}$ replaces the flat shine-through power here (moved to "Losses" below, where it belongs with the other loss channels).
- $P_{f,tot}, P_{f,th}, P_{f,b}, P_{f,bb}$ ("Pf") – the previously-empty 4th panel now shows $P_{f,bb}$ (Sec. "Beam-beam fusion" above – a flat zero while that's off), alongside the thermal and beam-target D-T/D-D totals; P_{useful} is still available from the Dashboard.
- $P_{\mathrm{shine}}, P_{\mathrm{orbit}}, P_{\mathrm{cx}}, P_{\mathrm{lost}}$ ("Losses") – the three loss channels (whichever CX-loss model is active feeds P_{cx}) plus their sum $P_{\mathrm{lost}} = P_{NB} - P_{\mathrm{useful}}$.
- $Y_{\mathrm{beam}}, Y_{\mathrm{thermal}}, Y_{\mathrm{bb}}, Y_{\mathrm{full}}$ ("Neutrons") – the neutron rate split by reaction channel: beam-target (D-T + D-D), thermal-thermal (D-T + D-D), beam-beam (Sec. "Beam-beam fusion" above – a flat zero while that's off), and the full total ($Y_{\mathrm{beam}} + Y_{\mathrm{thermal}} + Y_{\mathrm{bb}} = Y_{\mathrm{full}}$ exactly). Named Y, not R, so it isn't confused with the $R = u_{\mathrm{fast}}/U_{\mathrm{th}}$ scan group.

1.27 Important notation

- T_e: electron temperature
- T_i: thermal-ion temperature
- P_e: beam power deposited into electrons
- P_i: beam power deposited into ions
- L_e, L_i: beam-power electron / ion heating fractions ($L_e + L_i = 1$)
- P_{ie}: electron-ion energy exchange power
- P_{ECRH}, P_{ICRH}: ECRH and ICRH auxiliary heating power
- $P_{\mathrm{aux},e}$, $P_{\mathrm{aux},i}$: auxiliary heating to electrons / ions
- P_α, $P_{\alpha,e}$, $P_{\alpha,i}$: fusion alpha self-heating power (total, to electrons, to ions)
- $\tau_{E,e}$, $\tau_{E,i}$: electron / ion energy confinement times used in the balance
- n_{thermal}: thermal-ion density, $n_{\mathrm{sum}}(n_{e0}, Z_{\mathrm{eff}}) - n_{b0}$
- n_D, n_T: thermal deuterium and tritium densities
- n_{b0}: total fast-ion density

- ρ_{Li}: fast-ion toroidal-field Larmor radius at the injection energy
- Δr: passing drift-orbit radial width, $q_* \, \rho_{Li}$ (large-aspect model)
- ρ_θ: fast-ion poloidal gyroradius, $(B_t/\bar{B}_p) \, \rho_{Li}$ (ST orbit models)
- w_{pass}, w_{ban}: ST circulating / banana orbit widths
- f_t, f_c: trapped / circulating particle fractions (arbitrary A)
- q_a: arbitrary-aspect-ratio edge safety factor (Uckan form)
- f_{orbit}: first-orbit loss fraction of captured beam power
- P_{orbit}: first-orbit loss power
- $P_{f,tot}$, $P_{f,th}$, $P_{f,b}$: total, thermal, and beam-target fusion power (D-T)
- $P_{f,DT}$, $P_{f,DD}$: D-T and D-D fusion power ($P_{f,tot} = P_{f,DT} + P_{f,DD}$)
- R_n: neutron rate (14 MeV D-T + 2.45 MeV D-D branch)
- $\langle T \rangle$, T_0: volume-averaged and on-axis temperature (profile-corrected 0-D)
- p_n, p_{Te}, p_{Ti}: density and electron / ion temperature profile exponents ($(1-\rho^2)^{2p}$)
- τ_{IE}: effective electron-ion exchange time
- p_{th}, p_{fast}: thermal and fast-ion pressure
- β_t: toroidal beta
- f_{dil}: fast-ion dilution, $n_{b0}/(n_{\text{thermal}} + n_{b0})$
- τ_p^*: effective particle confinement time (incl. recycling)
- N_s: thermal inventory of species s, $\langle n_s \rangle V$
- $S_{\text{beam}\to s}$, $S_{\text{ext},s}$, $R_{\text{burn},s}$: beam, external fuelling and burn-up particle rates

1.28 Appendix A: confinement-scaling P definition – expected impact

This appendix expands on the caveat noted under "Confinement time" above. It is exploratory analysis, not a validated result: standalone scripts, outside the app and the solver, generated the three figures below (see `docs/Validation/tauE_scaling_Ploss_definition_scan.py` and `tauE_scaling_combined_comparison.py`). **Not yet acted on** – see the "confinement-scaling-p-loss-definition" note if revisiting this.

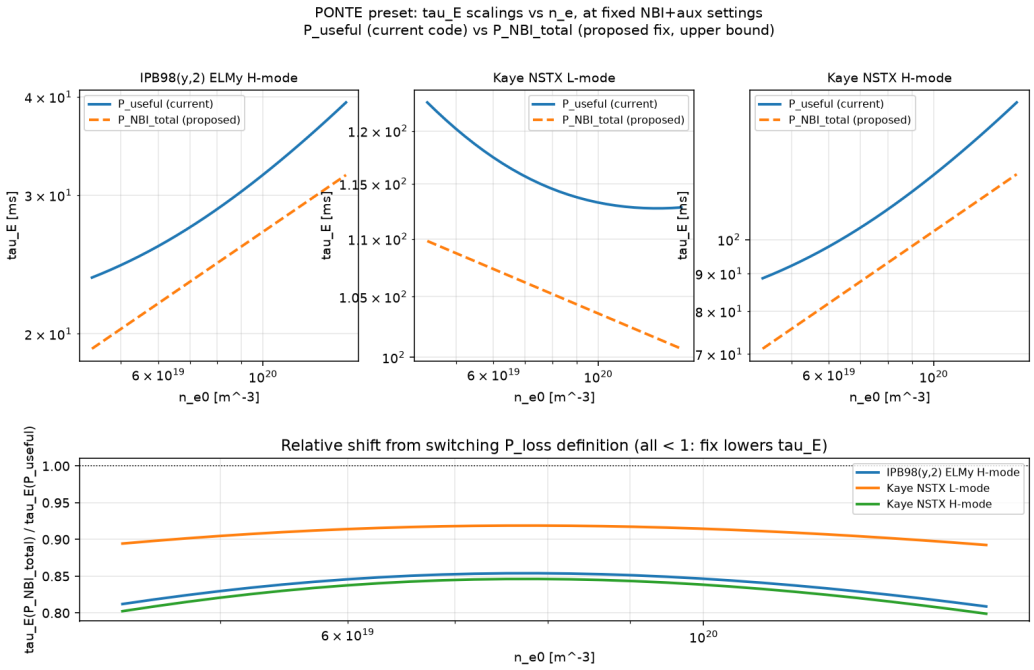
1.28.1 The question

IPB98(y,2) and the two Kaye NSTX scalings take a power argument, P. HI-Jass currently feeds them $P_{\text{useful}} + P_{\text{aux}}$ – NBI power *after* shine-through, first-orbit loss, and charge-exchange loss are all subtracted. The experimental databases these scalings were fit to use the coarser convention instead (absorbed power only, i.e. shine-through-corrected but *not* further reduced for orbit-loss/CX-loss of the already-absorbed fast-ion population). HI-Jass's convention therefore under-feeds P into these negative-exponent scalings, which should **overestimate** τ_E whenever orbit-loss/CX-loss are significant – worst for compact/ST-like geometries such as this project's own PONTE default.

1.28.2 PONTE: how much would the fix change τ_E?

For the PONTE preset, at fixed NBI/ECRH settings, switching from P_{useful} to the raw total injected NBI power (an upper bound on the proposed fix – shine-through would still be subtracted in the real fix) lowers τ_E by **˜10-20%**, depending on the scaling and the density:

Orbit-loss + CX-loss account for ˜25-32% of PONTE's injected NBI power over this scan – that gap is the direct cause of the shift above.



1.28.3 But the scaling choice itself matters far more

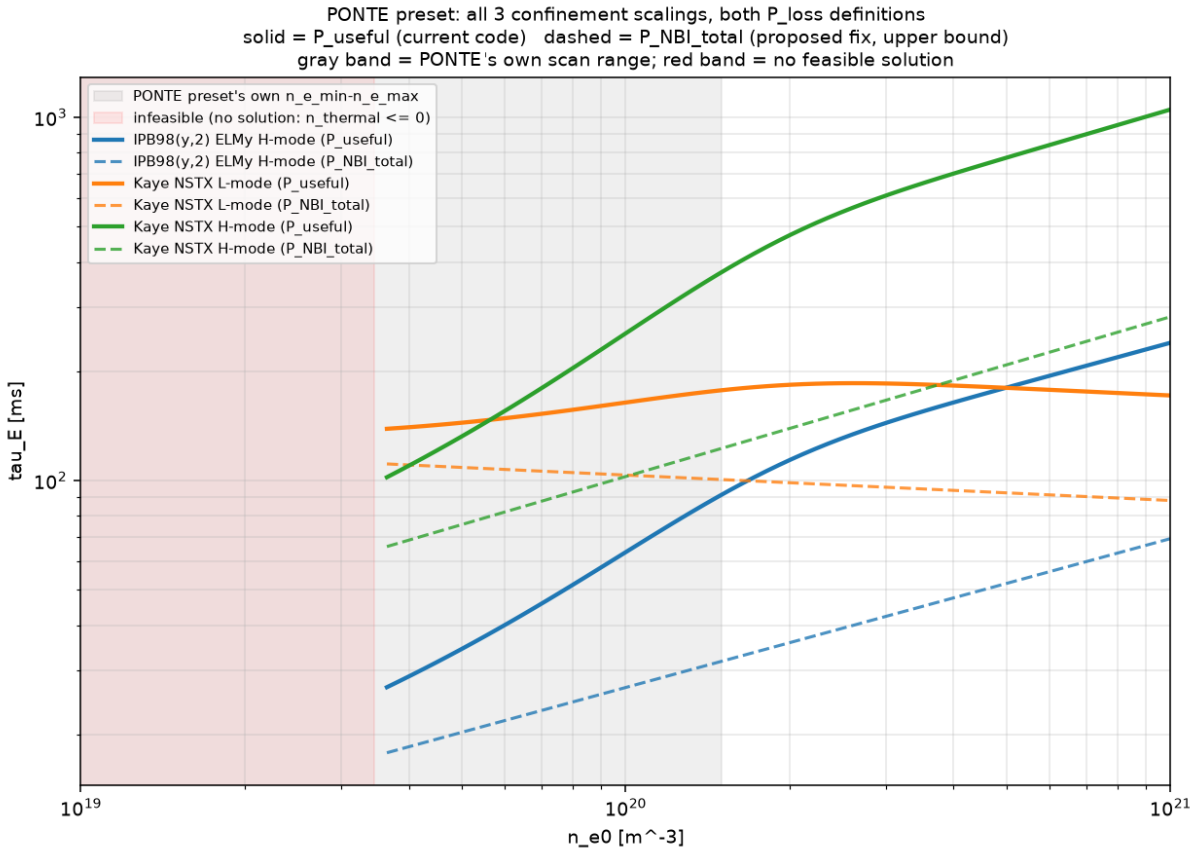
The same PONTE comparison, with all three scalings drawn on one axis, shows the **scaling-choice spread is ˜4-6x** across the density range – 20-30x larger than the P-definition effect above. Red shading marks densities where HI-Jass has no feasible solution at all for this preset (the model's thermal-ion density would go negative):

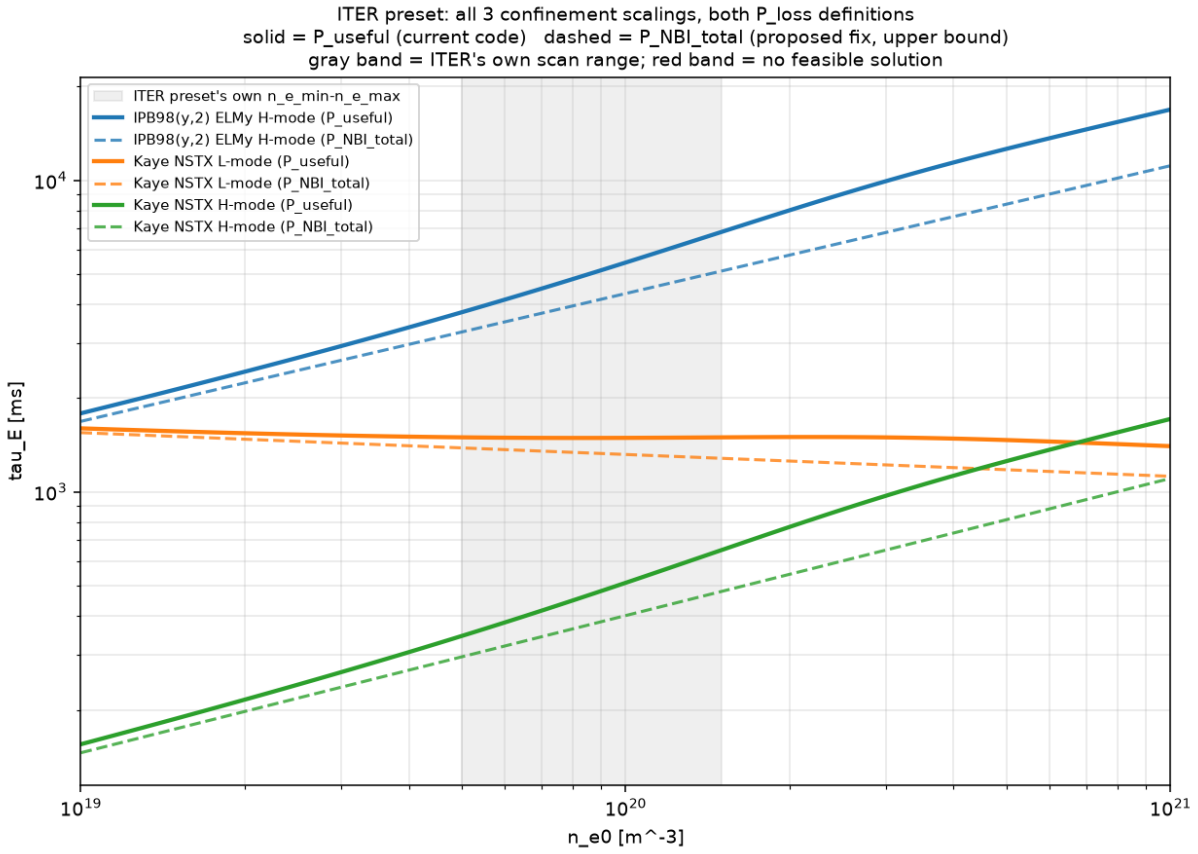
The same view for the **ITER** preset shows an even larger spread (˜10-12x), but with the *ranking reversed* – IPB98(y,2) now sits far above both Kaye curves, at multi-second τ_E values:

Read this ITER case with caution, not as competing physics estimates. Kaye's scalings were fit purely to NSTX data ($I_p \lesssim 1.5$ MA, $B_t \lesssim 0.5$ T); ITER's $I_p = 15$ MA / $B_t = 5.3$ T is a huge extrapolation *outside* that database, in the opposite direction from PONTE/NSTX-like devices. IPB98(y,2), by contrast, was fit including larger, more ITER-like devices, so it is the more trustworthy of the two here – the "spread" for ITER is mostly a statement about extrapolation validity, not a genuine disagreement in overlapping-validity physics.

1.28.4 Takeaway

Fixing the P_{loss} definition (sec. "Known limitation" above) is a real, worthwhile correction (˜10-20% on τ_E), but the choice of *which* scaling to use dominates the total uncertainty by an order of magnitude. Neither should be read as validated for a device far outside that scaling's own fitted dataset – check the Assumptions tab's low-aspect extrapolation flag, and prefer Kaye NSTX only for genuinely NSTX-like ($A \lesssim 1.6$, low B_t) geometries.


PONTE preset: all 3 confinement scalings, both P_loss definitions
solid = P_useful (current code) dashed = P_NBI_total (proposed fix, upper bound)
gray band = PONTE's own scan range; red band = no feasible solution
PONTE preset's own n_e_min-n_e_max
infeasible (no solution: n_thermal <= 0)
IPB98(y,2) ELMy H-mode (P_useful)
IPB98(y,2) ELMy H-mode (P_NBI_total)
Kaye NSTX L-mode (P_useful)
Kaye NSTX L-mode (P_NBI_total)
Kaye NSTX H-mode (P_useful)
Kaye NSTX H-mode (P_NBI_total)
tau_E [ms]
n_e0 [m^-3]


ITER preset: all 3 confinement scalings, both P_loss definitions
solid = P_useful (current code) dashed = P_NBI_total (proposed fix, upper bound)
gray band = ITER's own scan range; red band = no feasible solution
ITER preset's own n_e_min-n_e_max
IPB98(y,2) ELMy H-mode (P_useful)
IPB98(y,2) ELMy H-mode (P_NBI_total)
Kaye NSTX L-mode (P_useful)
Kaye NSTX L-mode (P_NBI_total)
Kaye NSTX H-mode (P_useful)
Kaye NSTX H-mode (P_NBI_total)
tau_E [ms]
n_e0 [m^-3]